Lora dotrenowana na cfg = 4, porónwanani z baseline i forward 4% na cfg=4

Metoda	CFG	LPIPS ↓	Corr. ↓	CLIP/Edit align. ↑
DDIM Inv.	4.0	0.525	0.172	0.725
Forward 4%	4.0	0.372	0.121	0.800
Shared LoRA	4.0	**0.1968**	**0.0955**	–

Lora uczona na kilku cfg
(2,3,4,5,7.5,10)
Inversja + rekonstrukcja na tym samym CFG

Shared - wspólna lora dla conditional i unconditional
Pair - osobnce lory dla conditional i unconditional

Branch loss - suma loss dla conditional i unconditional
Cfg loss - już złączony i przeskalowany output z conditional i unconditional

CFG	Wariant	MSE ↓	MAE ↓	PSNR ↑	SSIM ↑	CLIP Src. ↓	CLIP Tgt. ↑	Dir. CLIP ↑
2.0	DDIM Inv.	0.00827	0.0561	23.16	0.6563	33.49	74.08	34.08
	shared CFG loss	**0.00347**	**0.0341**	**26.09**	**0.8149**	30.14	80.20	**36.94**
	shared branch loss	0.00355	0.0349	25.61	0.8075	**29.67**	**80.47**	36.93
	pair CFG loss	0.02803	0.1352	15.70	0.5356	39.88	40.02	18.29
	pair branch loss	0.00531	0.0460	23.71	0.7347	29.85	79.80	36.79
3.0	DDIM Inv.	0.03172	0.1248	16.76	0.4171	35.83	72.24	32.61
	shared CFG loss	**0.00613**	**0.0427**	**24.49**	**0.7952**	29.60	81.94	37.81
	shared branch loss	0.00758	0.0503	22.76	0.7687	**28.80**	**82.36**	**37.97**
	pair CFG loss	0.02984	0.1374	15.44	0.5448	39.23	40.15	19.23
	pair branch loss	0.01657	0.0792	19.99	0.6474	30.06	80.47	36.98
4.0	DDIM Inv.	0.05467	0.1736	13.68	0.3464	36.72	67.66	30.82
	shared CFG loss	**0.01351**	**0.0658**	**21.40**	**0.7392**	**28.76**	82.58	38.40
	shared branch loss	0.01616	0.0764	19.77	0.6985	28.78	**83.37**	**38.60**
	pair CFG loss	0.03115	0.1373	15.26	0.5440	40.15	43.05	20.54
	pair branch loss	0.04813	0.1477	15.48	0.5134	30.88	77.53	36.03
5.0	DDIM Inv.	0.07277	0.2047	12.07	0.3355	36.02	64.94	30.04
	shared CFG loss	**0.02578**	**0.0977**	**18.49**	**0.6567**	29.48	81.98	38.13
	shared branch loss	0.03562	0.1221	16.33	0.5979	**29.24**	**82.74**	**38.28**
	pair CFG loss	0.03417	0.1413	14.89	0.5414	39.74	48.14	22.66
	pair branch loss	0.09919	0.2389	11.30	0.3538	33.69	69.37	32.19
7.5	DDIM Inv.	0.10139	0.2468	10.30	0.3163	37.10	60.04	28.53
	shared CFG loss	0.06526	0.1767	13.39	0.5101	**30.94**	**77.90**	**36.70**
	shared branch loss	0.10671	0.2462	10.62	0.3962	32.53	71.98	33.87
	pair CFG loss	**0.04216**	**0.1552**	**14.01**	**0.5401**	38.92	69.17	30.96
	pair branch loss	0.18254	0.3574	7.52	0.1869	37.73	51.68	24.42
10.0	DDIM Inv.	0.12182	0.2711	9.37	0.2946	37.99	55.59	26.80
	shared CFG loss	0.11570	0.2603	9.78	**0.3787**	**34.58**	68.99	**32.48**
	shared branch loss	0.15604	0.3131	8.37	0.2888	35.92	62.21	29.44
	pair CFG loss	**0.09063**	**0.2320**	**10.72**	0.3506	40.19	**70.83**	30.83
	pair branch loss	0.19553	0.3669	7.15	0.1688	37.75	45.89	22.54

Porównanie Lory uczonej na kilku guidance scale z uczoną na samym guidance scale = 4

Wariant	MSE ↓	MAE ↓	PSNR ↑	SSIM ↑
Single CFG = 4, shared CFG loss	**0.01108**	**0.05755**	**21.67**	**0.7582**
Mixed, shared CFG loss	0.01351	0.06584	21.40	0.7392
Mixed, shared branch loss	0.01616	0.07636	19.77	0.6985
Mixed, pair CFG loss	0.03115	0.13732	15.26	0.5440
Mixed, pair branch loss	0.04813	0.14771	15.48	0.5134

Lora i zwykłe DDIM inv, na guidance scale = 2 przy inwersji

CFG	Wariant	Kroki inv.	LoRA r/α	Corr. ↓	KL Div. ↓	MAE ↓	LPIPS ↓	PSNR ↑	SSIM ↑	CLIP Src. ↓	CLIP Tgt. ↑	Dir. CLIP ↑
5.0	DDIM Inv. (vanilla)	50	–	0.18093	2.42355	0.07241	0.18965	19.89	0.7444	32.59	80.50	37.17
	single-pass CFG	50	16/8	0.12247	0.21056	0.07168	0.18793	19.77	0.7482	29.83	84.28	39.03
	single-pass CFG	50	32/16	0.12193	0.19347	0.07128	0.18722	19.81	0.7492	29.97	84.36	39.02
	single-pass CFG	100	16/8	0.11678	0.02593	0.07139	0.19543	19.71	0.7422	29.88	**84.67**	39.04
	single-pass CFG	100	32/16	**0.11669**	**0.01876**	0.07110	0.19490	19.74	0.7427	29.81	84.62	39.03
	shared CFG loss	50	16/8	0.11746	0.03973	0.07021	0.18818	19.91	0.7468	29.90	84.60	39.08
	shared CFG loss	50	32/16	0.11685	0.06810	0.06720	0.18151	20.22	0.7571	30.02	84.65	**39.10**
	shared branch loss	50	16/8	0.12135	0.23944	0.06687	**0.17634**	**20.31**	**0.7612**	29.99	84.04	39.00
	shared branch loss	50	32/16	0.11975	0.13743	0.06731	0.17718	20.25	0.7598	**29.79**	84.22	39.10
	pair CFG loss	50	16/8	0.11895	0.13086	0.07507	0.21203	19.31	0.7261	29.91	84.23	38.93
	pair CFG loss	50	32/16	0.11842	0.06902	0.08103	0.22354	18.79	0.7107	30.07	84.65	38.99
	pair branch loss	50	16/8	0.12108	0.19438	0.06696	0.17797	20.27	0.7595	29.97	84.20	38.99
	pair branch loss	50	32/16	0.12084	0.18377	**0.06665**	0.17818	20.31	0.7598	29.97	84.15	38.99
7.5	DDIM Inv. (vanilla)	50	–	0.18670	2.85674	0.09246	0.22280	17.73	0.6931	32.45	81.76	37.97
	single-pass CFG	50	16/8	0.12402	0.26300	0.09034	0.21872	17.79	0.6975	30.04	85.04	39.62
	single-pass CFG	50	32/16	0.12347	0.24388	0.08989	0.21756	17.83	0.6986	30.07	85.13	39.63
	single-pass CFG	100	16/8	0.11721	0.03468	0.09050	0.22567	17.71	0.6927	29.71	**85.58**	39.78
	single-pass CFG	100	32/16	**0.11704**	**0.02567**	0.09011	0.22560	17.76	0.6930	**29.64**	85.44	**39.79**
	shared CFG loss	50	16/8	0.11798	0.04718	0.08976	0.22357	17.81	0.6918	30.37	85.36	39.59
	shared CFG loss	50	32/16	0.11715	0.07570	0.08645	0.21491	18.05	0.7039	30.26	85.54	39.66
	shared branch loss	50	16/8	0.12282	0.30069	0.08565	**0.20854**	**18.17**	**0.7095**	30.22	85.27	39.69
	shared branch loss	50	32/16	0.12097	0.18556	0.08655	0.21208	18.08	0.7063	30.16	85.23	39.70
	pair CFG loss	50	16/8	0.11960	0.13733	0.09411	0.24335	17.35	0.6748	30.20	85.23	39.61
	pair CFG loss	50	32/16	0.11905	0.07762	0.10063	0.25664	16.91	0.6569	30.07	85.42	39.62
	pair branch loss	50	16/8	0.12257	0.25404	0.08581	0.21113	18.13	0.7078	30.29	85.17	39.60
	pair branch loss	50	32/16	0.12222	0.23913	**0.08542**	0.21113	18.16	0.7084	30.22	85.15	39.61
10.0	DDIM Inv. (vanilla)	50	–	0.19018	3.12596	0.10830	0.25287	16.30	0.6473	32.73	82.12	38.16
	single-pass CFG	50	16/8	0.12523	0.30958	0.10503	0.24565	16.44	0.6507	30.23	85.40	39.84
	single-pass CFG	50	32/16	0.12459	0.28780	0.10442	0.24420	16.49	0.6526	30.19	85.51	39.88
	single-pass CFG	100	16/8	0.11721	0.04640	0.10525	0.25248	16.38	0.6471	30.08	86.05	39.99
	single-pass CFG	100	32/16	**0.11705**	**0.03532**	0.10500	0.25219	16.39	0.6481	**29.92**	**86.13**	**40.10**
	shared CFG loss	50	16/8	0.11833	0.05570	0.10489	0.25186	16.41	0.6448	30.70	85.75	39.84
	shared CFG loss	50	32/16	0.11729	0.08076	0.10212	0.24413	16.61	0.6553	30.38	85.63	39.85
	shared branch loss	50	16/8	0.12400	0.35233	0.10077	**0.23606**	**16.74**	**0.6620**	30.47	85.21	39.75
	shared branch loss	50	32/16	0.12171	0.22936	0.10142	0.23890	16.68	0.6595	30.50	85.29	39.75
	pair CFG loss	50	16/8	0.12022	0.14363	0.11060	0.27316	15.94	0.6267	30.59	85.57	39.79
	pair CFG loss	50	32/16	0.11960	0.08957	0.11708	0.28901	15.55	0.6089	30.60	85.73	39.74
	pair branch loss	50	16/8	0.12368	0.30700	0.10096	0.23881	16.71	0.6606	30.50	85.43	39.79
	pair branch loss	50	32/16	0.12320	0.28886	**0.10063**	0.23859	16.72	0.6611	30.49	85.40	39.75

CFG rek./ed.	Wariant	KL $\times 10^2 \downarrow$	Recon. MAE $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	CLIP Tgt. $\uparrow$	Dir. CLIP $\uparrow$
2.0	single-pass, train CFG 2, $r = 16/\alpha = 8$	0.13423	0.03559	0.11924	25.48	0.8182	80.81	36.79
	single-pass, train CFG 2, $r = 32/\alpha = 16$	**0.11196**	**0.03494**	**0.11560**	**25.61**	**0.8215**	**80.85**	**36.85**
	single-pass, train mixed CFG, $r = 16/\alpha = 8$	0.12721	0.05316	0.18844	22.49	0.7518	80.72	**36.85**
3.0	single-pass, train CFG 2, $r = 16/\alpha = 8$	0.16772	**0.04847**	0.14471	**22.95**	0.7977	82.91	38.13
	single-pass, train CFG 2, $r = 32/\alpha = 16$	**0.14265**	0.04848	**0.14460**	22.94	**0.7978**	**83.02**	**38.21**
	single-pass, train mixed CFG, $r = 16/\alpha = 8$	0.15528	0.05530	0.18650	21.76	0.7601	82.95	38.16
4.0	single-pass, train CFG 2, $r = 16/\alpha = 8$	0.19712	**0.06084**	**0.16806**	**21.10**	**0.7733**	83.81	38.68
	single-pass, train CFG 2, $r = 32/\alpha = 16$	**0.16978**	0.06113	0.16893	21.07	0.7728	83.77	38.66
	single-pass, train mixed CFG, $r = 16/\alpha = 8$	0.17898	0.06112	0.19842	20.72	0.7510	**84.05**	**38.78**
5.0	single-pass, train CFG 2, $r = 16/\alpha = 8$	0.21056	0.07168	0.18793	19.77	0.7482	84.28	39.03
	single-pass, train CFG 2, $r = 32/\alpha = 16$	**0.19347**	0.07128	**0.18722**	**19.81**	**0.7492**	84.36	39.02
	single-pass, train mixed CFG, $r = 16/\alpha = 8$	0.20023	**0.06815**	0.21270	19.78	0.7365	**84.56**	**39.07**
7.5	single-pass, train CFG 2, $r = 16/\alpha = 8$	0.26300	0.09034	0.21872	17.79	0.6975	85.04	39.62
	single-pass, train CFG 2, $r = 32/\alpha = 16$	**0.24388**	0.08989	**0.21756**	**17.83**	**0.6986**	85.13	39.63
	single-pass, train mixed CFG, $r = 16/\alpha = 8$	0.24658	**0.08589**	0.24852	17.77	0.6945	**85.31**	**39.76**
10.0	single-pass, train CFG 2, $r = 16/\alpha = 8$	0.30958	0.10503	0.24565	16.44	0.6507	85.40	39.84
	single-pass, train CFG 2, $r = 32/\alpha = 16$	**0.28780**	0.10442	**0.24420**	**16.49**	**0.6526**	85.51	39.88
	single-pass, train mixed CFG, $r = 16/\alpha = 8$	0.28899	**0.10215**	0.28166	16.32	0.6520	**85.77**	**40.01**

Inversja cfg = 2, to samo co na poprzednim porównanie najlepszych

CFG rek.	Metoda	MAE ↓	LPIPS ↓	PSNR ↑	SSIM ↑
5.0	DDIM Inv.	0.07130	0.18710	20.07	0.7450
	shared CFG	**0.06898**	**0.18266**	**20.13**	**0.7547**
7.5	DDIM Inv.	0.09263	0.22367	17.79	0.6899
	shared CFG	**0.08653**	**0.21072**	**18.21**	**0.7068**
10.0	DDIM Inv.	0.10770	0.25394	16.38	0.6429
	shared CFG	**0.10139**	**0.23854**	**16.80**	**0.6585**

Wyniki PnP inversion dla SD1.5
CFG = 1

Porównanie edycji metodą PnP

Method		Structure	Background Preservation				CLIP Similarity	
Inverse	**Editing**	**Distance**$_{\times 10^3}$ ↓	**PSNR** ↑	**LPIPS**$_{\times 10^3}$ ↓	**MSE**$_{\times 10^4}$ ↓	**SSIM**$_{\times 10^2}$ ↑	**Whole** ↑	**Edited** ↑
DDIM	**PnP***	28.22	22.28	113.46	83.64	79.05	25.41	22.55
Direct Inversion	**PnP***	**24.29**	**22.46**	**106.06**	**80.45**	**79.68**	25.41	22.62
LoRA $r = 8,\ \alpha = 4$	**PnP**	27.35	22.29	114.42	83.25	79.43	25.64	22.72
LoRA $r = 16,\ \alpha = 8$	**PnP**	27.18	22.32	114.16	82.44	79.46	**25.65**	22.78
LoRA $r = 32,\ \alpha = 16$	**PnP**	27.08	22.33	113.47	81.70	79.54	25.61	**22.81**

W następnych slajdach przykłady dla r32

Change object (oryginalny obraz, rekonstrukcja, edycja)

source prompt: fishes in the ocean
target prompt: sharks in the ocean



source prompt: a cute little bunny with big eyes
target prompt: a cute little pig with big eyes



Add object

source prompt: a round painting of a forest with flowers, trees
target prompt: a round painting of a forest with deer, flowers, trees



source prompt: a dog wearing space suit
target prompt: a dog wearing space suit with flowers in mouth



Delete object

source prompt: a bee flies over a flowering tree branch
target prompt: a flowering tree branch



source prompt: a husky dog running on a path in the woods
target prompt: a path in the woods



Change attribute color

source prompt: A white shirt
target prompt: A yellow shirt



source prompt: a black bird with a yellow beak and yellow feet
target prompt: a green bird with a yellow beak and yellow feet



Method		Structure	Background Preservation				CLIP Similarity	
Inverse	**Editing**	**Distance**$_{\times 10^3}$ ↓	**PSNR** ↑	**LPIPS**$_{\times 10^3}$ ↓	**MSE**$_{\times 10^4}$ ↓	**SSIM**$_{\times 10^2}$ ↑	**Whole** ↑	**Edited** ↑
DDIM	**P2P**	69.43	17.87	208.80	219.88	71.14	25.01	22.44
NT†	**P2P**	13.44	27.03	60.67	35.86	84.11	24.75	21.86
NP	**P2P**	16.17	26.21	69.01	39.73	83.40	24.61	21.87
StyleD	**P2P**	**11.65**	26.05	66.10	38.63	83.42	24.78	21.72
Direct Inversion	**P2P**	**11.65**	**27.22**	**54.55**	**32.86**	**84.76**	25.02	22.10
LoRA $r = 8,\ \alpha = 4$	**P2P**	70.45	17.81	211.10	221.32	71.37	**25.38**	**22.57**
LoRA $r = 16,\ \alpha = 8$	**P2P**	70.40	17.82	210.98	220.66	71.37	25.36	22.57
LoRA $r = 32,\ \alpha = 16$	**P2P**	70.47	17.82	210.95	220.22	71.39	25.36	22.55
DDIM	**MasaCtrl**	28.38	22.17	106.62	86.97	79.67	23.96	21.16
Direct Inversion	**MasaCtrl**	**24.70**	**22.64**	**87.94**	**81.09**	**81.33**	**24.38**	**21.35**
LoRA $r = 8,\ \alpha = 4$	**MasaCtrl**	26.56	22.25	106.53	87.91	80.29	24.16	21.12
LoRA $r = 16,\ \alpha = 8$	**MasaCtrl**	26.49	22.29	105.30	87.57	80.37	24.22	21.18
LoRA $r = 32,\ \alpha = 16$	**MasaCtrl**	26.67	22.28	105.38	87.27	80.45	24.13	21.18
DDIM	**P2P-Zero**	61.68	20.44	172.22	144.12	74.67	22.80	20.54
Direct Inversion	**P2P-Zero**	**49.22**	**21.53**	**138.98**	**127.32**	**77.05**	23.31	**21.05**
LoRA $r = 8,\ \alpha = 4$	**P2P-Zero**	51.59	21.33	145.87	140.59	76.98	**23.46**	20.99
LoRA $r = 16,\ \alpha = 8$	**P2P-Zero**	52.39	21.26	146.89	142.57	76.90	23.35	21.05
LoRA $r = 32,\ \alpha = 16$	**P2P-Zero**	51.65	21.29	147.02	140.93	77.03	23.41	21.04
DDIM	**PnP***	28.22	22.28	113.46	83.64	79.05	25.41	22.55
Direct Inversion	**PnP***	**24.29**	**22.46**	**106.06**	**80.45**	**79.68**	25.41	22.62
LoRA $r = 8,\ \alpha = 4$	**PnP**	27.35	22.29	114.42	83.25	79.43	25.64	22.72
LoRA $r = 16,\ \alpha = 8$	**PnP**	27.18	22.32	114.16	82.44	79.46	**25.65**	22.78
LoRA $r = 32,\ \alpha = 16$	**PnP**	27.08	22.33	113.47	81.70	79.54	25.61	**22.81**

Reprodukcja wyników z pnp_inversion

Editor	Inversion	Source	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP whole ↑	CLIP edited ↑
P2P	DDIM	Original	**69.43**	17.87	208.80	219.88	71.14	25.01	22.44
P2P	DDIM	Reproduced	70.54	**17.87**	208.84	**219.35**	**71.64**	**25.28**	**22.56**
P2P	Direct Inversion	Original	11.65	27.22	54.55	32.86	84.76	25.02	22.10
P2P	Direct Inversion	Reproduced	**11.01**	**27.27**	**54.01**	**32.60**	**85.32**	**25.20**	**22.14**
MasaCtrl	DDIM	Original	28.38	22.17	106.62	86.97	79.67	23.96	21.16
MasaCtrl	DDIM	Reproduced	**27.09**	**22.19**	**105.34**	**86.35**	**80.32**	**24.23**	**21.20**
MasaCtrl	Direct Inversion	Original	24.70	22.64	87.94	81.09	81.33	24.38	21.35
MasaCtrl	Direct Inversion	Reproduced	**23.70**	**22.67**	**87.85**	**80.85**	**81.85**	**24.60**	**21.46**
P2P-Zero	DDIM	Original	61.68	20.44	**172.22**	144.12	74.67	22.80	20.54
P2P-Zero	DDIM	Reproduced	**61.25**	**20.47**	172.26	**143.71**	**75.39**	**22.93**	**20.74**
P2P-Zero	Direct Inversion	Original	49.22	21.53	138.98	**127.32**	77.05	23.31	21.05
P2P-Zero	Direct Inversion	Reproduced	**49.09**	**21.57**	**138.39**	129.77	**77.73**	**23.47**	**21.08**
PnP	DDIM	Original	28.22	22.28	113.46	83.64	79.05	25.41	22.55
PnP	DDIM	Reproduced	**27.33**	**22.32**	**112.73**	**82.94**	**79.58**	**25.58**	**22.57**
PnP	Direct Inversion	Original	24.29	22.46	106.06	80.45	79.68	25.41	22.62
PnP	Direct Inversion	Reproduced	**23.35**	**22.46**	**105.49**	**79.93**	**80.22**	**25.64**	**22.67**

Wyniki dla DirectInversion połączonego z LoRĄ

Editor	Inversion / rank	α	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP target (edit) ↑
PnP	Original DDIM	–	28.22	22.28	113.46	83.64	79.05	25.41	22.55
PnP	Original DirectInv	–	24.29	**22.46**	**106.06**	**80.45**	79.68	25.41	22.62
PnP	r8	4	27.35 ± 25.20	22.29 ± 4.45	114.42 ± 64.99	83.25 ± 60.37	79.43 ± 12.89	25.64 ± 3.72	22.72 ± 4.79
PnP	r16	8	27.18 ± 24.68	22.32 ± 4.51	114.16 ± 64.64	82.44 ± 59.26	79.45 ± 12.87	25.65 ± 3.74	22.78 ± 4.81
PnP	r32	16	27.08 ± 24.68	22.33 ± 4.28	113.47 ± 64.74	81.70 ± 58.70	79.54 ± 12.85	25.61 ± 3.76	22.81 ± 4.84
PnP + DirectInv	r8	4	24.45 ± 20.52	22.38 ± 4.54	110.48 ± 62.69	81.77 ± 60.05	79.85 ± 12.80	**25.81** ±3.71	22.85 ± 4.85
PnP + DirectInv	r16	8	24.28 ± 20.21	22.40 ± 4.55	110.24 ± 62.55	81.56 ± 59.99	79.88 ± 12.80	25.80 ± 3.73	**22.86** ±4.86
PnP + DirectInv	r32	16	**24.24** ±20.14	22.40 ± 4.53	110.08 ± 62.43	81.43 ± 59.91	**79.90** ±12.79	25.80 ± 3.72	22.84 ± 4.88
P2P	Original DDIM	–	69.43	17.87	208.80	219.88	71.14	25.01	22.44
P2P	Original DirectInv	–	**11.65**	**27.22**	**54.55**	**32.86**	**84.76**	25.02	22.10
P2P	r8	4	70.45 ± 39.60	17.81 ± 3.92	211.10 ± 106.52	221.32 ± 145.76	71.37 ± 15.86	**25.38** ±3.64	**22.57** ±4.53
P2P	r16	8	70.40 ± 38.77	17.82 ± 3.91	210.98 ± 106.11	220.66 ± 144.99	71.37 ± 15.79	25.36 ± 3.62	22.57 ± 4.58
P2P	r32	16	70.47 ± 40.18	17.82 ± 3.90	210.95 ± 106.10	220.22 ± 145.22	71.39 ± 15.87	25.36 ± 3.59	22.55 ± 4.52
P2P + DirectInv	r8	4	14.46 ± 11.87	25.96 ± 5.43	73.96 ± 51.91	44.87 ± 46.73	83.40 ± 12.24	25.27 ± 3.87	22.33 ± 4.91
P2P + DirectInv	r16	8	14.30 ± 11.91	25.95 ± 5.37	73.20 ± 50.93	44.76 ± 46.29	83.48 ± 12.19	25.31 ± 3.82	22.39 ± 4.84
P2P + DirectInv	r32	16	14.30 ± 11.78	25.95 ± 5.22	72.31 ± 48.55	43.62 ± 43.56	83.50 ± 12.12	25.29 ± 3.91	22.35 ± 4.82
MasaCtrl	Original DDIM	–	28.38	22.17	106.62	86.97	79.67	23.96	21.16
MasaCtrl	Original DirectInv	–	**24.70**	**22.64**	**87.94**	**81.09**	**81.33**	24.38	21.35
MasaCtrl	r8	4	26.56 ± 19.33	22.25 ± 4.69	106.53 ± 61.47	87.91 ± 70.68	80.28 ± 12.74	24.16 ± 3.80	21.12 ± 4.67
MasaCtrl	r16	8	26.49 ± 19.75	22.29 ± 4.71	105.30 ± 60.89	87.57 ± 72.82	80.37 ± 12.77	24.22 ± 3.80	21.18 ± 4.65
MasaCtrl	r32	16	26.67 ± 19.94	22.28 ± 4.69	105.38 ± 61.60	87.27 ± 72.33	80.45 ± 12.65	24.13 ± 3.79	21.18 ± 4.68
MasaCtrl + DirectInv	r8	4	28.48 ± 21.49	21.89 ± 4.87	98.63 ± 58.63	99.08 ± 85.44	80.49 ± 12.72	**24.64** ±3.65	**21.50** ±4.66
MasaCtrl + DirectInv	r16	8	28.50 ± 21.47	21.89 ± 4.90	98.89 ± 58.64	99.89 ± 88.99	80.47 ± 12.72	24.62 ± 3.65	21.49 ± 4.67
MasaCtrl + DirectInv	r32	16	28.37 ± 21.07	21.89 ± 4.84	98.78 ± 58.20	98.68 ± 85.10	80.50 ± 12.68	24.60 ± 3.60	21.49 ± 4.65
Pix2Pix-Zero	Original DDIM	–	61.68	20.44	172.22	144.12	74.67	22.80	20.54
Pix2Pix-Zero	Original DirectInv	–	**49.22**	**21.53**	**138.98**	**127.32**	**77.05**	23.31	21.05
Pix2Pix-Zero	r8	4	51.59 ± 51.82	21.33 ± 5.58	145.87 ± 119.77	140.59 ± 169.23	76.98 ± 15.83	23.46 ± 4.28	20.99 ± 4.70
Pix2Pix-Zero	r16	8	52.38 ± 53.04	21.26 ± 5.55	146.89 ± 119.91	142.57 ± 177.53	76.90 ± 15.84	23.35 ± 4.34	21.05 ± 4.70
Pix2Pix-Zero	r32	16	51.65 ± 52.88	21.29 ± 5.55	147.02 ± 120.50	140.93 ± 170.82	77.03 ± 15.82	23.41 ± 4.44	21.04 ± 4.75
Pix2Pix-Zero + DirectInv	r8	4	51.42 ± 49.89	21.27 ± 5.53	146.53 ± 118.70	139.61 ± 161.66	77.02 ± 15.69	23.36 ± 4.34	20.99 ± 4.68
Pix2Pix-Zero + DirectInv	r16	8	51.62 ± 50.63	21.23 ± 5.48	147.11 ± 120.67	140.85 ± 170.35	77.04 ± 15.68	**23.49** ±4.28	21.02 ± 4.78
Pix2Pix-Zero + DirectInv	r32	16	53.69 ± 55.20	21.16 ± 5.57	149.48 ± 121.79	144.91 ± 171.78	76.68 ± 15.96	23.46 ± 4.35	**21.11** ±4.80

Transfer Stylu

Inversion Method	CLIP Prompt Alignment ↑	Set Consistency (DINO) ↑	Set Consistency (CSD) ↑	Style Similarity (DINO) ↑	Style Similarity (CSD) ↑
DDIM	0.8151 (ref.)	0.4849 (ref.)	0.6559 (ref.)	0.4521 (ref.)	0.5326 (ref.)
DDIM forward 4%	0.8123 (-0.0028)	0.4897 (+0.0048)	0.6611 (+0.0052)	0.4452 (-0.0069)	0.5305 (-0.0021)
Single-pass $r = 16$	0.8105 (-0.0046)	0.4812 (-0.0037)	0.6586 (+0.0027)	0.4513 (-0.0008)	0.5328 (+0.0002)
Single-pass $r = 32$	0.8121 (-0.0030)	0.4807 (-0.0042)	0.6589 (+0.0030)	0.4507 (-0.0014)	0.5343 (+0.0017)
Pair branch loss $r = 16$	0.8156 (+0.0005)	0.4798 (-0.0051)	0.6638 (+0.0079)	0.4471 (-0.0050)	0.5362 (+0.0036)
Pair branch loss $r = 32$	0.8152 (+0.0001)	0.4799 (-0.0050)	0.6637 (+0.0078)	0.4479 (-0.0042)	0.5362 (+0.0036)
Pair CFG loss $r = 16$	0.7833 (-0.0318)	0.5711 (+0.0862)	**0.6887 (+0.0328)**	**0.4870 (+0.0349)**	**0.5506 (+0.0180)**
Pair CFG loss $r = 32$	0.7779 (-0.0372)	**0.6087 (+0.1238)**	0.6867 (+0.0308)	0.4522 (+0.0001)	0.5033 (-0.0293)
Shared branch loss $r = 16$	0.8176 (+0.0025)	0.4778 (-0.0071)	0.6614 (+0.0055)	0.4447 (-0.0074)	0.5350 (+0.0024)
Shared branch loss $r = 32$	0.8149 (-0.0002)	0.4828 (-0.0021)	0.6637 (+0.0078)	0.4512 (-0.0009)	0.5399 (+0.0073)
Shared CFG loss $r = 16$	0.8228 (+0.0077)	0.4709 (-0.0140)	0.6477 (-0.0082)	0.4386 (-0.0135)	0.5162 (-0.0164)
Shared CFG loss $r = 32$	**0.8232 (+0.0081)**	0.4693 (-0.0156)	0.6479 (-0.0080)	0.4362 (-0.0159)	0.5141 (-0.0185)

CFG inwersji 2, CFG generacji 10, offsetem inwersji 5, shared score shift 2.

Inversion Method	CLIP Prompt Alignment ↑	Set Consistency (DINO) ↑	Set Consistency (CSD) ↑	Style Similarity (DINO) ↑	Style Similarity (CSD) ↑
DDIM	0.8110 (ref.)	0.4883 (ref.)	0.6597 (ref.)	0.4558 (ref.)	0.5345 (ref.)
DDIM forward 4%	0.8054 (-0.0056)	0.4936 (+0.0053)	0.6665 (+0.0068)	0.4576 (+0.0018)	0.5420 (+0.0075)
No-CFG loss $r = 16$	0.8102 (-0.0008)	0.4885 (+0.0002)	0.6609 (+0.0012)	0.4552 (-0.0006)	0.5308 (-0.0037)
No-CFG loss $r = 32$	0.8106 (-0.0004)	0.4953 (+0.0070)	0.6612 (+0.0015)	0.4628 (+0.0070)	0.5340 (-0.0005)
Single-pass $r = 16$	0.8105 (-0.0005)	0.4812 (-0.0071)	0.6586 (-0.0011)	0.4513 (-0.0045)	0.5328 (-0.0017)
Single-pass $r = 32$	0.8121 (+0.0011)	0.4807 (-0.0076)	0.6589 (-0.0008)	0.4507 (-0.0051)	0.5343 (-0.0002)
Pair branch loss $r = 16$	0.8068 (-0.0042)	0.4822 (-0.0061)	0.6613 (+0.0016)	0.4489 (-0.0069)	0.5342 (-0.0003)
Pair branch loss $r = 32$	0.8088 (-0.0022)	0.4801 (-0.0082)	0.6588 (-0.0009)	0.4481 (-0.0077)	0.5335 (-0.0010)
Pair CFG loss $r = 16$	0.8061 (-0.0049)	**0.5129 (+0.0246)**	**0.6690 (+0.0093)**	**0.4877 (+0.0319)**	**0.5729 (+0.0384)**
Pair CFG loss $r = 32$	0.8107 (-0.0003)	0.5056 (+0.0173)	0.6623 (+0.0026)	0.4830 (+0.0272)	0.5653 (+0.0308)
Shared branch loss $r = 16$	0.8095 (-0.0015)	0.4825 (-0.0058)	0.6589 (-0.0008)	0.4509 (-0.0049)	0.5337 (-0.0008)
Shared branch loss $r = 32$	0.8102 (-0.0008)	0.4822 (-0.0061)	0.6597 (+0.0000)	0.4512 (-0.0046)	0.5379 (+0.0034)
Shared CFG loss $r = 16$	**0.8123 (+0.0013)**	0.4819 (-0.0064)	0.6543 (-0.0054)	0.4491 (-0.0067)	0.5230 (-0.0115)
Shared CFG loss $r = 32$	0.8117 (+0.0007)	0.4833 (-0.0050)	0.6564 (-0.0033)	0.4492 (-0.0066)	0.5221 (-0.0124)

CFG inwersji 1, CFG generacji 10, offsetem inwersji 5, shared score shift 2.

Na kolejnych slide wszysktie używają CFG inwersji = 1

Metoda	CLIP prompt ↑	DINO set ↑	CSD set ↑	DINO style ↑	CSD style ↑
Offset = 0, **shared score shift** = 1, **CFG generowania** = 8					
DDIM	0.8591 (ref.)	0.3576 (ref.)	0.5698 (ref.)	0.3505 (ref.)	0.4337 (ref.)
DDIM + forward 4%	0.8607 (+0.0016)	0.3667 (+0.0091)	0.5646 (-0.0053)	0.3577 (+0.0072)	0.4426 (+0.0089)
No-CFG	0.8608 (+0.0017)	0.3551 (-0.0025)	0.5613 (-0.0085)	0.3525 (+0.0019)	0.4322 (-0.0016)
Single-pass	**0.8637 (+0.0046)**	0.3577 (+0.0001)	0.5683 (-0.0016)	0.3532 (+0.0027)	0.4372 (+0.0035)
Pair branch	0.8602 (+0.0011)	0.3592 (+0.0016)	0.5682 (-0.0016)	0.3532 (+0.0027)	0.4338 (+0.0000)
Pair CFG	0.8576 (-0.0015)	**0.3831 (+0.0255)**	**0.5763 (+0.0065)**	**0.3673 (+0.0168)**	**0.4561 (+0.0224)**
Shared branch	0.8618 (+0.0027)	0.3590 (+0.0014)	0.5685 (-0.0014)	0.3533 (+0.0028)	0.4326 (-0.0012)
Shared CFG	0.8619 (+0.0028)	0.3575 (-0.0001)	0.5677 (-0.0021)	0.3435 (-0.0070)	0.4207 (-0.0130)
Offset = 3, **shared score shift** = 1, **CFG generowania** = 8					
DDIM	0.8589 (ref.)	0.3588 (ref.)	0.5689 (ref.)	0.3508 (ref.)	0.4347 (ref.)
DDIM + forward 4%	0.8618 (+0.0029)	0.3670 (+0.0082)	0.5677 (-0.0012)	0.3588 (+0.0080)	0.4449 (+0.0102)
No-CFG	0.8607 (+0.0017)	0.3572 (-0.0016)	0.5626 (-0.0063)	0.3543 (+0.0035)	0.4331 (-0.0016)
Single-pass	**0.8632 (+0.0043)**	0.3588 (+0.0000)	0.5685 (-0.0003)	0.3534 (+0.0026)	0.4393 (+0.0046)
Pair branch	0.8627 (+0.0038)	0.3592 (+0.0004)	0.5684 (-0.0005)	0.3531 (+0.0022)	0.4355 (+0.0007)
Pair CFG	0.8564 (-0.0026)	**0.3841 (+0.0253)**	**0.5793 (+0.0105)**	**0.3715 (+0.0206)**	**0.4601 (+0.0253)**
Shared branch	0.8613 (+0.0024)	0.3579 (-0.0009)	0.5672 (-0.0016)	0.3529 (+0.0020)	0.4357 (+0.0009)
Shared CFG	0.8585 (-0.0004)	0.3575 (-0.0013)	0.5681 (-0.0008)	0.3442 (-0.0067)	0.4190 (-0.0157)
Offset = 5, **shared score shift** = 1, **CFG generowania** = 8					
DDIM	0.8559 (ref.)	0.3678 (ref.)	0.5708 (ref.)	0.3579 (ref.)	0.4380 (ref.)
DDIM + forward 4%	0.8578 (+0.0019)	0.3685 (+0.0007)	0.5681 (-0.0028)	0.3606 (+0.0027)	0.4443 (+0.0063)
No-CFG	0.8580 (+0.0021)	0.3622 (-0.0056)	0.5665 (-0.0044)	0.3606 (+0.0027)	0.4426 (+0.0046)
Single-pass	**0.8607 (+0.0048)**	0.3642 (-0.0036)	0.5728 (+0.0020)	0.3667 (+0.0088)	0.4484 (+0.0104)
Pair branch	0.8595 (+0.0036)	0.3644 (-0.0034)	0.5707 (-0.0001)	0.3595 (+0.0016)	0.4446 (+0.0066)
Pair CFG	0.8555 (-0.0005)	**0.3905 (+0.0227)**	**0.5798 (+0.0089)**	**0.3784 (+0.0205)**	**0.4641 (+0.0261)**
Shared branch	0.8594 (+0.0035)	0.3623 (-0.0055)	0.5744 (+0.0035)	0.3583 (+0.0004)	0.4433 (+0.0053)
Shared CFG	0.8561 (+0.0001)	0.3598 (-0.0081)	0.5682 (-0.0027)	0.3472 (-0.0107)	0.4203 (-0.0177)
Offset = 5, **shared score shift** = 3, **CFG generowania** = 8					
DDIM	0.7511 (ref.)	0.6166 (ref.)	0.7643 (ref.)	0.5535 (ref.)	0.6238 (ref.)
DDIM + forward 4%	0.7460 (-0.0051)	0.6216 (+0.0051)	0.7721 (+0.0078)	0.5497 (-0.0038)	0.6195 (-0.0043)
No-CFG	0.7495 (-0.0016)	0.6092 (-0.0073)	0.7671 (+0.0028)	0.5486 (-0.0049)	0.6147 (-0.0091)
Single-pass	0.7514 (+0.0003)	0.6146 (-0.0020)	0.7709 (+0.0066)	0.5537 (+0.0002)	0.6197 (-0.0041)
Pair branch	0.7545 (+0.0034)	0.5992 (-0.0173)	0.7628 (-0.0015)	0.5425 (-0.0110)	0.6143 (-0.0095)
Pair CFG	0.7477 (-0.0034)	**0.6280 (+0.0114)**	**0.7726 (+0.0082)**	**0.5935 (+0.0399)**	**0.6711 (+0.0473)**
Shared branch	0.7523 (+0.0012)	0.6019 (-0.0146)	0.7621 (-0.0022)	0.5450 (-0.0086)	0.6173 (-0.0066)
Shared CFG	**0.7567 (+0.0055)**	0.6003 (-0.0162)	0.7620 (-0.0023)	0.5474 (-0.0062)	0.6171 (-0.0068)

Offset = 0, **shared score shift** = 2, **CFG generowania** = 8					
DDIM	0.7996 (ref.)	0.5080 (ref.)	0.6873 (ref.)	0.4793 (ref.)	0.5504 (ref.)
DDIM + forward 4%	0.7976 (-0.0020)	0.5108 (+0.0028)	**0.6959 (+0.0087)**	0.4738 (-0.0055)	0.5579 (+0.0075)
No-CFG	0.8006 (+0.0010)	0.4953 (-0.0127)	0.6849 (-0.0024)	0.4737 (-0.0056)	0.5453 (-0.0050)
Single-pass	0.7989 (-0.0007)	0.4950 (-0.0129)	0.6845 (-0.0027)	0.4744 (-0.0049)	0.5478 (-0.0025)
Pair branch	0.8003 (+0.0007)	0.4912 (-0.0168)	0.6805 (-0.0068)	0.4685 (-0.0109)	0.5420 (-0.0084)
Pair CFG	0.8059 (+0.0063)	**0.5111 (+0.0031)**	0.6911 (+0.0039)	**0.4958 (+0.0165)**	**0.5876 (+0.0372)**
Shared branch	0.7977 (-0.0019)	0.4946 (-0.0133)	0.6855 (-0.0018)	0.4719 (-0.0074)	0.5460 (-0.0043)
Shared CFG	**0.8067 (+0.0071)**	0.4986 (-0.0094)	0.6839 (-0.0034)	0.4726 (-0.0067)	0.5435 (-0.0069)
Offset = 3, **shared score shift** = 2, **CFG generowania** = 8					
DDIM	0.7989 (ref.)	0.5047 (ref.)	0.6880 (ref.)	0.4788 (ref.)	0.5535 (ref.)
DDIM + forward 4%	0.7977 (-0.0012)	0.5096 (+0.0049)	**0.6982 (+0.0102)**	0.4738 (-0.0050)	0.5567 (+0.0032)
No-CFG	0.7969 (-0.0020)	0.4986 (-0.0061)	0.6835 (-0.0045)	0.4767 (-0.0021)	0.5437 (-0.0098)
Single-pass	0.8006 (+0.0017)	0.4939 (-0.0108)	0.6851 (-0.0029)	0.4750 (-0.0038)	0.5498 (-0.0037)
Pair branch	0.8013 (+0.0024)	0.4913 (-0.0134)	0.6823 (-0.0058)	0.4684 (-0.0104)	0.5448 (-0.0087)
Pair CFG	0.8020 (+0.0031)	**0.5248 (+0.0201)**	0.6960 (+0.0079)	**0.5045 (+0.0256)**	**0.5945 (+0.0410)**
Shared branch	0.7983 (-0.0007)	0.4914 (-0.0133)	0.6851 (-0.0030)	0.4691 (-0.0098)	0.5461 (-0.0074)
Shared CFG	**0.8067 (+0.0078)**	0.4953 (-0.0094)	0.6807 (-0.0073)	0.4726 (-0.0062)	0.5434 (-0.0101)
Offset = 4, **shared score shift** = 2, **CFG generowania** = 8					
DDIM	0.7986 (ref.)	0.5081 (ref.)	0.6885 (ref.)	0.4816 (ref.)	0.5560 (ref.)
DDIM + forward 4%	0.7972 (-0.0015)	0.5132 (+0.0051)	0.6973 (+0.0088)	0.4765 (-0.0051)	0.5578 (+0.0018)
No-CFG	0.7983 (-0.0004)	0.4981 (-0.0100)	0.6840 (-0.0045)	0.4781 (-0.0036)	0.5485 (-0.0075)
Single-pass	0.7990 (+0.0003)	0.4957 (-0.0125)	0.6874 (-0.0011)	0.4775 (-0.0042)	0.5547 (-0.0013)
Pair branch	0.7994 (+0.0008)	0.4898 (-0.0183)	0.6827 (-0.0058)	0.4681 (-0.0135)	0.5467 (-0.0093)
Pair CFG	0.7995 (+0.0009)	**0.5294 (+0.0213)**	**0.7011 (+0.0126)**	**0.5103 (+0.0287)**	**0.6003 (+0.0443)**
Shared branch	0.7984 (-0.0002)	0.4916 (-0.0165)	0.6829 (-0.0056)	0.4698 (-0.0118)	0.5471 (-0.0089)
Shared CFG	**0.8071 (+0.0085)**	0.4950 (-0.0131)	0.6805 (-0.0080)	0.4736 (-0.0080)	0.5433 (-0.0127)
Offset = 5, **shared score shift** = 2, **CFG generowania** = 8					
DDIM	0.7977 (ref.)	0.5092 (ref.)	0.6902 (ref.)	0.4838 (ref.)	0.5601 (ref.)
DDIM + forward 4%	0.7953 (-0.0024)	0.5152 (+0.0059)	0.6973 (+0.0071)	0.4813 (-0.0025)	0.5618 (+0.0017)
No-CFG	0.7975 (-0.0001)	0.5007 (-0.0086)	0.6858 (-0.0044)	0.4804 (-0.0034)	0.5518 (-0.0083)
Single-pass	0.8001 (+0.0025)	0.4984 (-0.0108)	0.6907 (+0.0005)	0.4800 (-0.0038)	0.5547 (-0.0054)
Pair branch	0.7994 (+0.0017)	0.4956 (-0.0137)	0.6840 (-0.0061)	0.4714 (-0.0124)	0.5495 (-0.0106)
Pair CFG	0.7967 (-0.0010)	**0.5351 (+0.0259)**	**0.7029 (+0.0127)**	**0.5177 (+0.0338)**	**0.6023 (+0.0422)**
Shared branch	0.7984 (+0.0007)	0.4969 (-0.0124)	0.6856 (-0.0046)	0.4737 (-0.0101)	0.5519 (-0.0082)
Shared CFG	**0.8077 (+0.0101)**	0.4969 (-0.0123)	0.6815 (-0.0087)	0.4748 (-0.0091)	0.5462 (-0.0139)

Offset = 0, **shared score shift** = 4, **CFG generowania** = 8

Metoda	CLIP prompt ↑	DINO set ↑	CSD set ↑	DINO style ↑	CSD style ↑
DDIM	0.7322 (ref.)	0.6630 (ref.)	0.8004 (ref.)	0.5805 (ref.)	0.6470 (ref.)
DDIM + forward 4%	0.7161 (-0.0161)	**0.6890 (+0.0260)**	**0.8112 (+0.0108)**	0.5894 (+0.0090)	0.6523 (+0.0053)
No-CFG	0.7330 (+0.0008)	0.6601 (-0.0029)	0.8016 (+0.0012)	0.5765 (-0.0039)	0.6356 (-0.0114)
Single-pass	0.7287 (-0.0035)	0.6610 (-0.0020)	0.7998 (-0.0006)	0.5803 (-0.0002)	0.6390 (-0.0080)
Pair branch	0.7278 (-0.0045)	0.6542 (-0.0088)	0.7982 (-0.0022)	0.5757 (-0.0047)	0.6393 (-0.0077)
Pair CFG	0.7233 (-0.0090)	0.6577 (-0.0053)	0.7995 (-0.0009)	**0.6067 (+0.0262)**	**0.6890 (+0.0421)**
Shared branch	0.7323 (+0.0001)	0.6510 (-0.0120)	0.7991 (-0.0013)	0.5744 (-0.0060)	0.6410 (-0.0059)
Shared CFG	**0.7364 (+0.0042)**	0.6413 (-0.0217)	0.7937 (-0.0067)	0.5791 (-0.0014)	0.6456 (-0.0014)

Offset = 3, **shared score shift** = 4, **CFG generowania** = 8

Metoda	CLIP prompt ↑	DINO set ↑	CSD set ↑	DINO style ↑	CSD style ↑
DDIM	0.7278 (ref.)	0.6611 (ref.)	0.8001 (ref.)	0.5787 (ref.)	0.6485 (ref.)
DDIM + forward 4%	0.7167 (-0.0111)	**0.6838 (+0.0227)**	**0.8095 (+0.0094)**	0.5831 (+0.0044)	0.6512 (+0.0027)
No-CFG	0.7283 (+0.0005)	0.6644 (+0.0033)	0.8040 (+0.0039)	0.5812 (+0.0024)	0.6417 (-0.0068)
Single-pass	0.7283 (+0.0006)	0.6646 (+0.0035)	0.8048 (+0.0047)	0.5806 (+0.0019)	0.6441 (-0.0043)
Pair branch	0.7272 (-0.0006)	0.6616 (+0.0005)	0.8039 (+0.0039)	0.5806 (+0.0019)	0.6439 (-0.0045)
Pair CFG	0.7207 (-0.0070)	0.6665 (+0.0054)	0.8035 (+0.0034)	**0.6132 (+0.0344)**	**0.6934 (+0.0450)**
Shared branch	0.7265 (-0.0013)	0.6602 (-0.0009)	0.8012 (+0.0011)	0.5802 (+0.0014)	0.6462 (-0.0022)
Shared CFG	**0.7345 (+0.0067)**	0.6426 (-0.0185)	0.7937 (-0.0063)	0.5768 (-0.0020)	0.6475 (-0.0010)

Offset = 5, **shared score shift** = 4, **CFG generowania** = 8

Metoda	CLIP prompt ↑	DINO set ↑	CSD set ↑	DINO style ↑	CSD style ↑
DDIM	0.7223 (ref.)	0.6719 (ref.)	0.8063 (ref.)	0.5887 (ref.)	0.6560 (ref.)
DDIM + forward 4%	0.7123 (-0.0100)	**0.6881 (+0.0162)**	**0.8135 (+0.0072)**	0.5904 (+0.0017)	0.6530 (-0.0030)
No-CFG	0.7241 (+0.0018)	0.6633 (-0.0085)	0.8038 (-0.0025)	0.5823 (-0.0064)	0.6482 (-0.0078)
Single-pass	0.7266 (+0.0043)	0.6666 (-0.0053)	0.8040 (-0.0023)	0.5877 (-0.0010)	0.6496 (-0.0064)
Pair branch	0.7284 (+0.0061)	0.6588 (-0.0131)	0.8019 (-0.0044)	0.5792 (-0.0095)	0.6457 (-0.0103)
Pair CFG	0.7161 (-0.0062)	0.6782 (+0.0064)	0.8131 (+0.0068)	**0.6276 (+0.0389)**	**0.7059 (+0.0499)**
Shared branch	0.7267 (+0.0044)	0.6587 (-0.0131)	0.8023 (-0.0041)	0.5808 (-0.0079)	0.6483 (-0.0077)
Shared CFG	**0.7337 (+0.0114)**	0.6514 (-0.0205)	0.8002 (-0.0061)	0.5817 (-0.0070)	0.6531 (-0.0029)

Offset = 3, **shared score shift** = 2, **CFG generowania** = 10

Metoda	CLIP prompt ↑	DINO set ↑	CSD set ↑	DINO style ↑	CSD style ↑
DDIM	0.8071 (ref.)	0.4900 (ref.)	0.6602 (ref.)	0.4576 (ref.)	0.5316 (ref.)
DDIM + forward 4%	0.8068 (-0.0002)	0.4969 (+0.0069)	**0.6759 (+0.0157)**	0.4568 (-0.0008)	0.5416 (+0.0101)
No-CFG	0.8030 (-0.0041)	0.4879 (-0.0021)	0.6592 (-0.0010)	0.4561 (-0.0015)	0.5282 (-0.0034)
Single-pass	0.8084 (+0.0013)	0.4820 (-0.0080)	0.6622 (+0.0020)	0.4552 (-0.0024)	0.5325 (+0.0010)
Pair branch	0.8040 (-0.0030)	0.4816 (-0.0084)	0.6627 (+0.0025)	0.4465 (-0.0111)	0.5272 (-0.0044)
Pair CFG	0.8093 (+0.0023)	**0.5039 (+0.0139)**	0.6703 (+0.0101)	**0.4783 (+0.0207)**	**0.5682 (+0.0366)**
Shared branch	0.8071 (+0.0001)	0.4838 (-0.0062)	0.6608 (+0.0006)	0.4475 (-0.0101)	0.5280 (-0.0035)
Shared CFG	**0.8137 (+0.0066)**	0.4807 (-0.0093)	0.6574 (-0.0028)	0.4506 (-0.0070)	0.5215 (-0.0101)

Offset = 4, **shared score shift** = 2, **CFG generowania** = 10

Metoda	CLIP prompt ↑	DINO set ↑	CSD set ↑	DINO style ↑	CSD style ↑
DDIM	0.8030 (ref.)	0.4942 (ref.)	0.6637 (ref.)	0.4605 (ref.)	0.5345 (ref.)
DDIM + forward 4%	0.8061 (+0.0031)	0.4983 (+0.0041)	0.6726 (+0.0088)	0.4563 (-0.0042)	0.5426 (+0.0080)
No-CFG	0.8027 (-0.0003)	0.4889 (-0.0054)	0.6628 (-0.0009)	0.4573 (-0.0033)	0.5298 (-0.0047)
Single-pass	0.8067 (+0.0037)	0.4820 (-0.0122)	0.6639 (+0.0002)	0.4555 (-0.0050)	0.5345 (+0.0000)
Pair branch	0.8052 (+0.0022)	0.4832 (-0.0110)	0.6650 (+0.0012)	0.4486 (-0.0119)	0.5312 (-0.0033)
Pair CFG	0.8064 (+0.0035)	**0.5131 (+0.0189)**	**0.6756 (+0.0119)**	**0.4857 (+0.0252)**	**0.5744 (+0.0399)**
Shared branch	0.8066 (+0.0036)	0.4830 (-0.0113)	0.6617 (-0.0021)	0.4487 (-0.0119)	0.5304 (-0.0041)
Shared CFG	**0.8135 (+0.0105)**	0.4840 (-0.0102)	0.6580 (-0.0057)	0.4507 (-0.0098)	0.5228 (-0.0118)

Offset = 5, **shared score shift** = 2, **CFG generowania** = 10

Metoda	CLIP prompt ↑	DINO set ↑	CSD set ↑	DINO style ↑	CSD style ↑
DDIM	0.8037 (ref.)	0.4973 (ref.)	0.6671 (ref.)	0.4636 (ref.)	0.5383 (ref.)
DDIM + forward 4%	0.8057 (+0.0020)	0.5012 (+0.0039)	0.6744 (+0.0074)	0.4605 (-0.0031)	0.5429 (+0.0046)
No-CFG	0.8032 (-0.0005)	0.4926 (-0.0047)	0.6650 (-0.0020)	0.4621 (-0.0015)	0.5337 (-0.0046)
Single-pass	0.8073 (+0.0036)	0.4825 (-0.0148)	0.6657 (-0.0014)	0.4559 (-0.0077)	0.5364 (-0.0018)
Pair branch	0.8042 (+0.0005)	0.4849 (-0.0124)	0.6662 (-0.0009)	0.4524 (-0.0113)	0.5339 (-0.0044)
Pair CFG	0.8057 (+0.0020)	**0.5192 (+0.0219)**	**0.6803 (+0.0133)**	**0.4923 (+0.0287)**	**0.5793 (+0.0411)**
Shared branch	0.8041 (+0.0004)	0.4892 (-0.0081)	0.6661 (-0.0009)	0.4551 (-0.0085)	0.5344 (-0.0039)
Shared CFG	**0.8137 (+0.0100)**	0.4862 (-0.0111)	0.6585 (-0.0086)	0.4528 (-0.0108)	0.5249 (-0.0133)

Reference style



ddim



Ddim + 4% foward



Pair branch



Pair cfg



Shared branch



Shared cfg



Single pass



A baby penguin in watercolor painting style

Style-Rank

Pipeline	Inwersja / ranga	α	ImageReward ↑	CLIP-Text ↑	CLIP-Image ↑	DINOv2 ↑	CSD style ↑
StyleAligned	DDIM (inv. CFG 1)	–	$\mathbf{-1.029 \pm 0.779}$	19.428 ± 3.178	73.324 ± 9.138	39.437 ± 9.977	0.6148 ± 0.1507
StyleAligned	Pair CFG LoRA $r = 16$ (inv. CFG 1)	8	-1.043 ± 0.765	$\mathbf{19.737 \pm 2.975}$	72.526 ± 9.042	$\mathbf{39.718 \pm 9.772}$	0.6101 ± 0.1473
StyleAligned	No-CFG LoRA $r = 16$ (inv. CFG 1)	8	-1.077 ± 0.787	19.434 ± 3.174	$\mathbf{73.630 \pm 8.664}$	39.157 ± 9.939	$\mathbf{0.6169 \pm 0.1513}$
StyleAligned	No-CFG LoRA $r = 32$ (inv. CFG 1)	16	-1.089 ± 0.762	19.366 ± 3.227	73.392 ± 8.858	39.294 ± 9.999	0.6137 ± 0.1512
Inversion-InstantStyle	DDIM	–	-1.071 ± 0.871	19.691 ± 3.258	$\mathbf{73.518 \pm 10.220}$	$\mathbf{43.969 \pm 16.461}$	$\mathbf{0.6368 \pm 0.1518}$
Inversion-InstantStyle	No-CFG LoRA $r = 16$	8	-1.044 ± 0.853	19.745 ± 3.116	73.172 ± 10.184	43.774 ± 16.341	0.6340 ± 0.1521
Inversion-InstantStyle	No-CFG LoRA $r = 32$	16	$\mathbf{-1.025 \pm 0.848}$	$\mathbf{19.779 \pm 3.110}$	73.325 ± 10.057	43.379 ± 16.339	0.6307 ± 0.1529

Metoda	Źródło	ImageReward ↑	CLIP-Text ↑	CLIP-Image ↑	DINOv2 ↑
StyleAligned	Oryginał	-1.260	19.260	68.720	36.290
	Reprodukcja	$\mathbf{-1.029}$ $(\Delta_A + 0.231)$	$\mathbf{19.428}$ $(\Delta_A + 0.168)$	$\mathbf{73.324}$ $(\Delta_A + 4.604)$	$\mathbf{39.437}$ $(\Delta_A + 3.147)$
VisualStyle	Oryginał	-0.720	22.120	$\mathbf{66.680}$	$\mathbf{20.800}$
	Reprodukcja	$\mathbf{-0.489}$ $(\Delta_A + 0.231)$	$\mathbf{22.347}$ $(\Delta_A + 0.227)$	63.745 $(\Delta_A - 2.935)$	19.749 $(\Delta_A - 1.051)$
IP-Adapter	Oryginał	-2.030	15.010	$\mathbf{83.660}$	$\mathbf{40.500}$
	Reprodukcja	$\mathbf{-1.934}$ $(\Delta_A + 0.096)$	$\mathbf{15.548}$ $(\Delta_A + 0.538)$	82.909 $(\Delta_A - 0.751)$	40.460 $(\Delta_A - 0.040)$
StyleShot	Oryginał	$\mathbf{-0.380}$	21.340	$\mathbf{65.040}$	$\mathbf{23.040}$
	Reprodukcja	-0.418 $(\Delta_A - 0.038)$	$\mathbf{21.730}$ $(\Delta_A + 0.390)$	63.839 $(\Delta_A - 1.201)$	22.049 $(\Delta_A - 0.991)$
CSGO	Oryginał	$\mathbf{-0.290}$	22.160	$\mathbf{61.730}$	16.850
	Reprodukcja	-0.313 $(\Delta_A - 0.023)$	$\mathbf{22.268}$ $(\Delta_A + 0.108)$	59.749 $(\Delta_A - 1.981)$	$\mathbf{16.879}$ $(\Delta_A + 0.029)$
InstantStyle	Oryginał	-0.130	22.780	$\mathbf{66.430}$	18.480
	Reprodukcja	$\mathbf{0.043}$ $(\Delta_A + 0.173)$	$\mathbf{23.214}$ $(\Delta_A + 0.434)$	64.910 $(\Delta_A - 1.520)$	$\mathbf{18.955}$ $(\Delta_A + 0.475)$
Inversion-InstantStyle	Oryginał	-1.300	18.900	$\mathbf{76.600}$	$\mathbf{49.420}$
	Reprodukcja	$\mathbf{-1.071}$ $(\Delta_A + 0.229)$	$\mathbf{19.691}$ $(\Delta_A + 0.791)$	73.518 $(\Delta_A - 3.082)$	43.969 $(\Delta_A - 5.451)$

StyleBench (StyleShot)

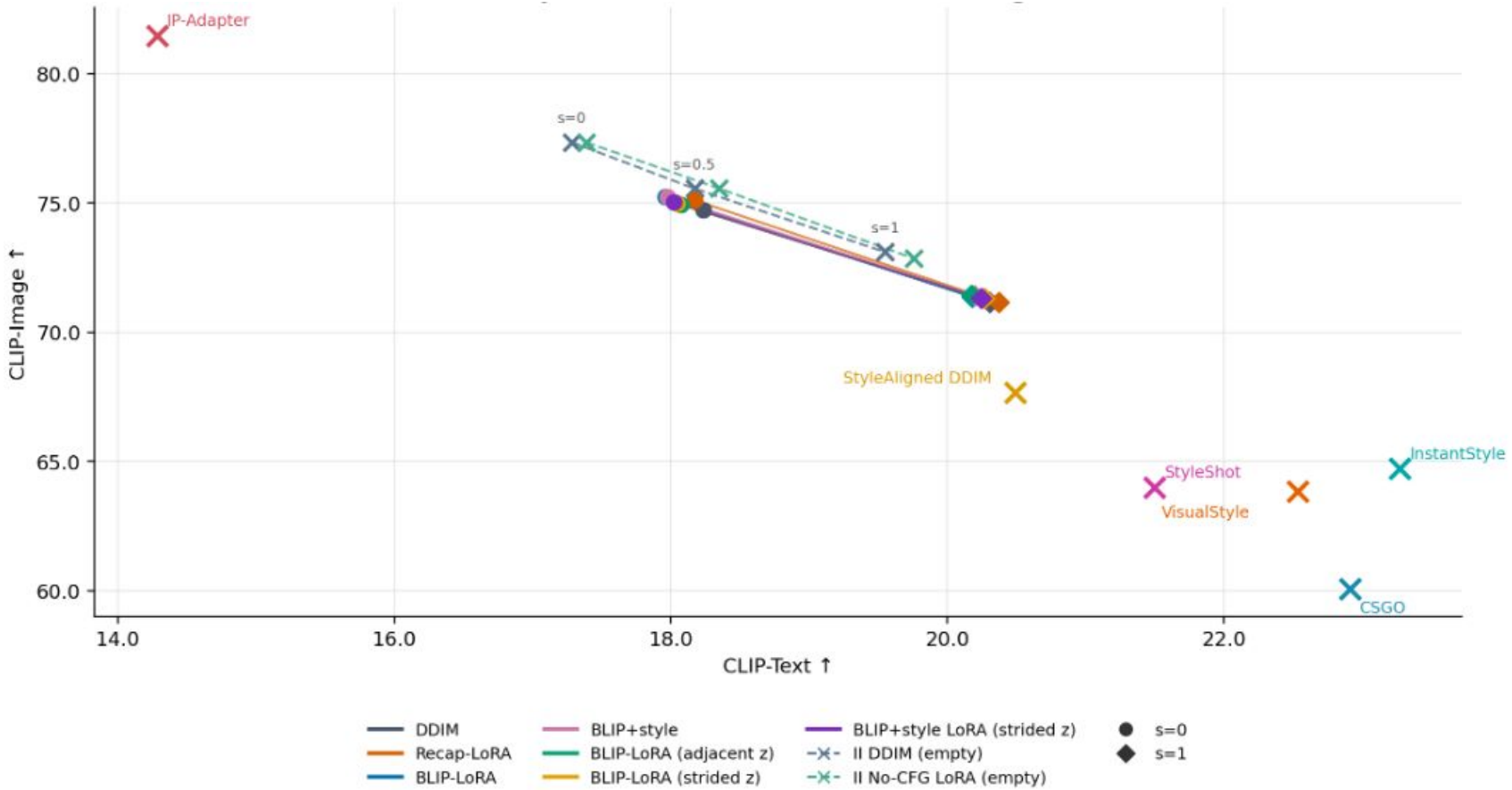
Pipeline	Inwersja / ranga	α	ImageReward ↑	CLIP-Text ↑	CLIP-Image ↑	DINOv2 ↑	CSD style ↑
StyleAligned	DDIM (inv. CFG 1)	–	**−1.218 ± 0.573**	19.698 ± 2.306	69.704 ± 7.557	32.950 ± 7.117	0.5501 ± 0.1292
StyleAligned	Pair CFG LoRA $r = 16$ (inv. CFG 1)	8	−1.249 ± 0.564	**19.809 ± 2.238**	69.541 ± 7.696	**33.628 ± 7.528**	**0.5784 ± 0.1306**
StyleAligned	No-CFG LoRA $r = 16$ (inv. CFG 1)	8	−1.243 ± 0.573	19.702 ± 2.370	70.018 ± 7.698	32.871 ± 7.143	0.5556 ± 0.1329
StyleAligned	No-CFG LoRA $r = 32$ (inv. CFG 1)	16	−1.243 ± 0.572	19.701 ± 2.386	**70.031 ± 7.698**	32.871 ± 7.194	0.5552 ± 0.1331

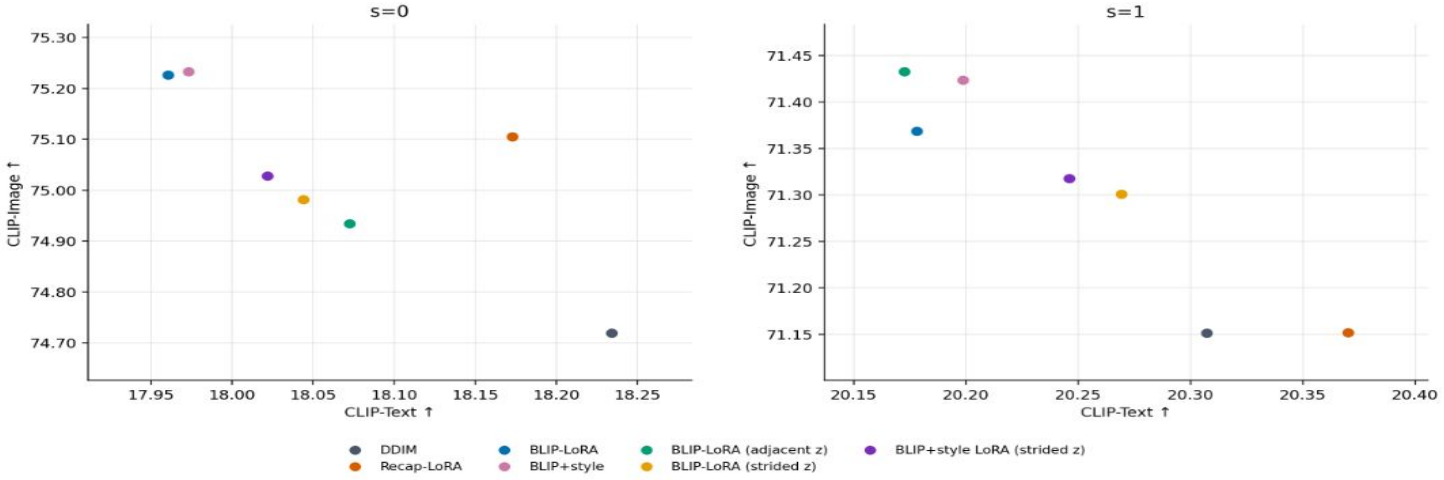
Table 1: Quantitative comparison from human preference and clip scoring on text and image alignment with SOTA text-driven style transfer methods. Best result is marked in **bold**.

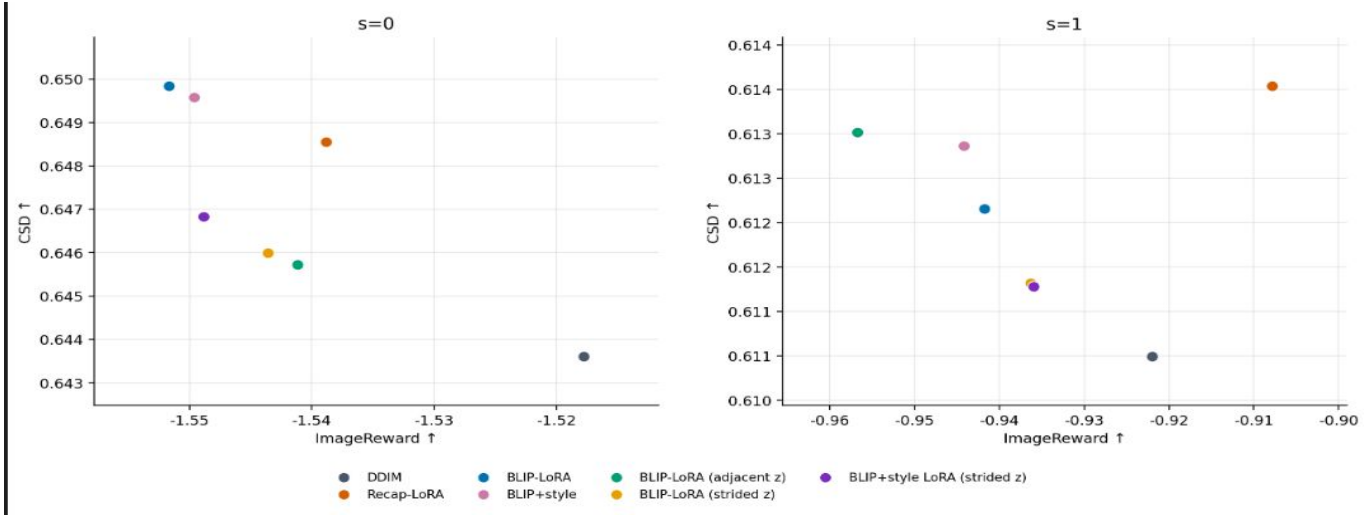
Human	StyleCrafter	DEADifff	StyleDrop	InST	StyleAligned	StyleShot
text ↑	9.7%	19.3%	6.0%	12.7%	8.0%	**44.3%**
image ↑	14.3%	8.0%	4.0%	6.3%	17.3%	**50.0%**

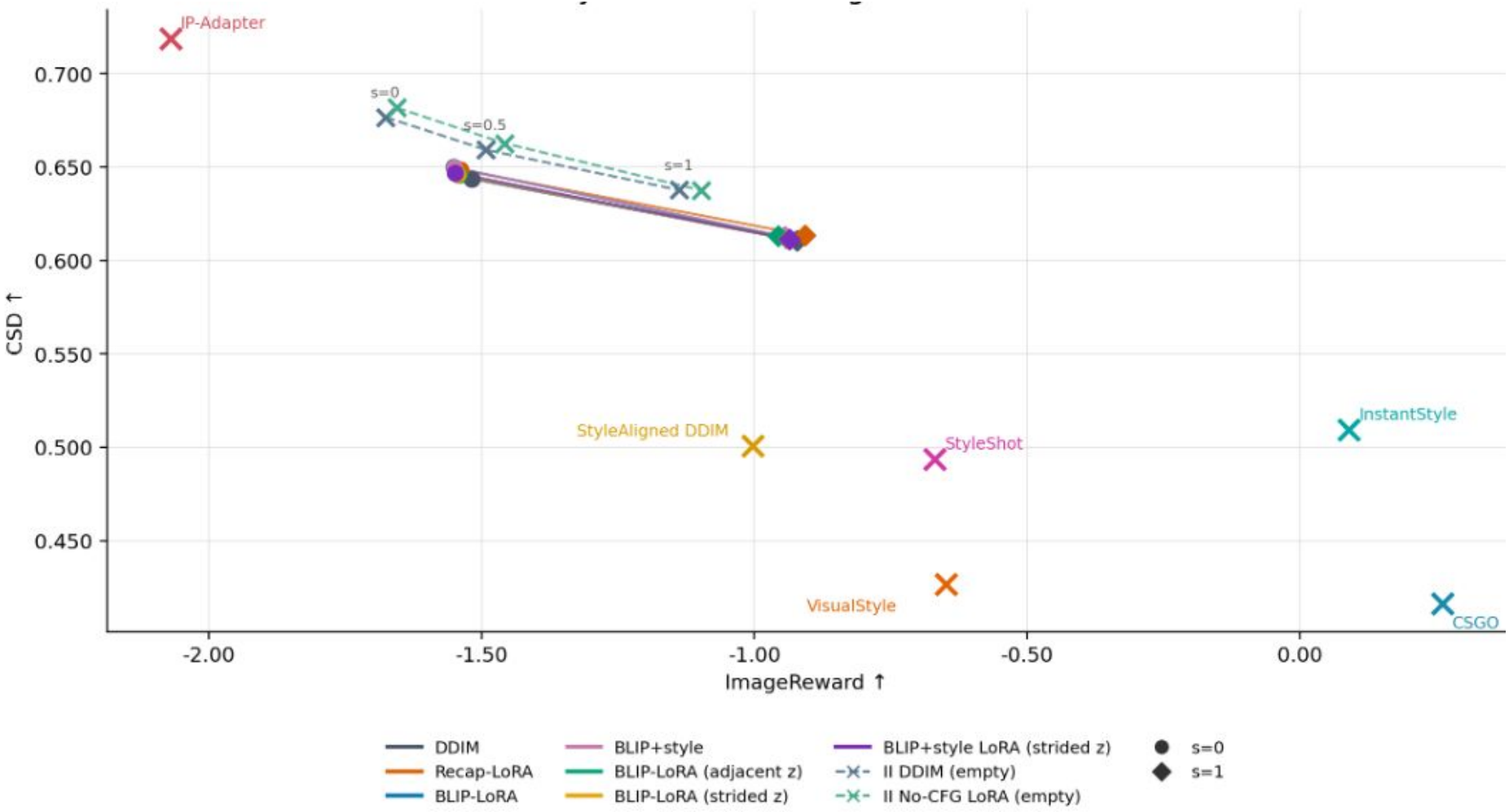
CLIP	StyleCrafter	DEADifff	StyleDrop	InST	StyleAligned	StyleShot
text ↑	0.202	**0.232**	0.220	0.204	0.213	0.219
image ↑	**0.706**	0.597	0.621	0.623	0.680	0.640

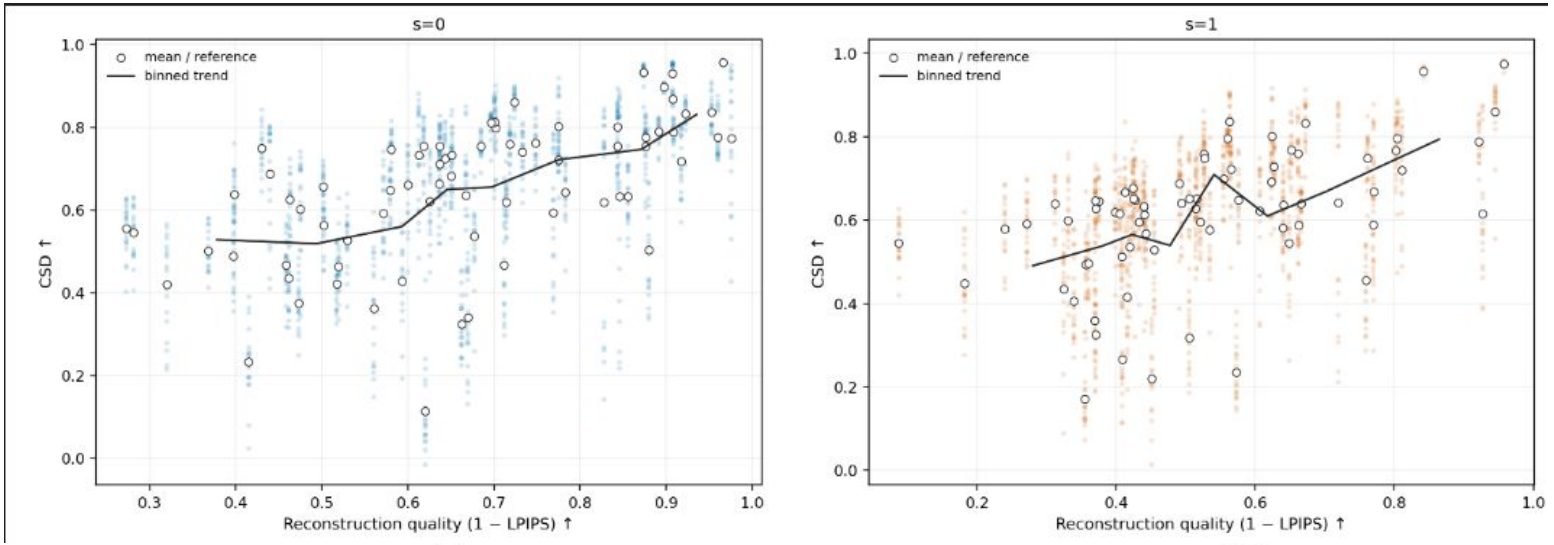
StyleBench

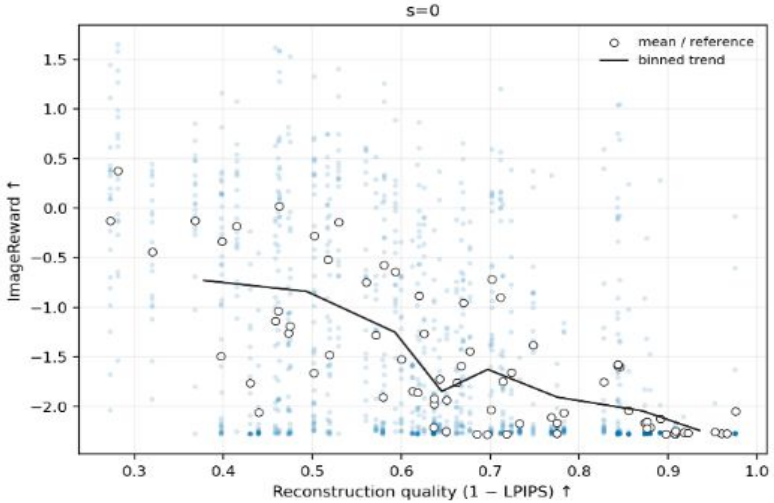

IP-Adapter
s=0
s=0.5
s=1
StyleAligned DDIM
StyleShot
InstantStyle
VisualStyle
CSGO
80.0
75.0
70.0
65.0
60.0
CLIP-Image ↑
14.0
16.0
18.0
20.0
22.0
CLIP-Text ↑
DDIM
Recap-LoRA
BLIP-LoRA
BLIP+style
BLIP-LoRA (adjacent z)
BLIP-LoRA (strided z)
BLIP+style LoRA (strided z)
II DDIM (empty)
II No-CFG LoRA (empty)
s=0
s=1

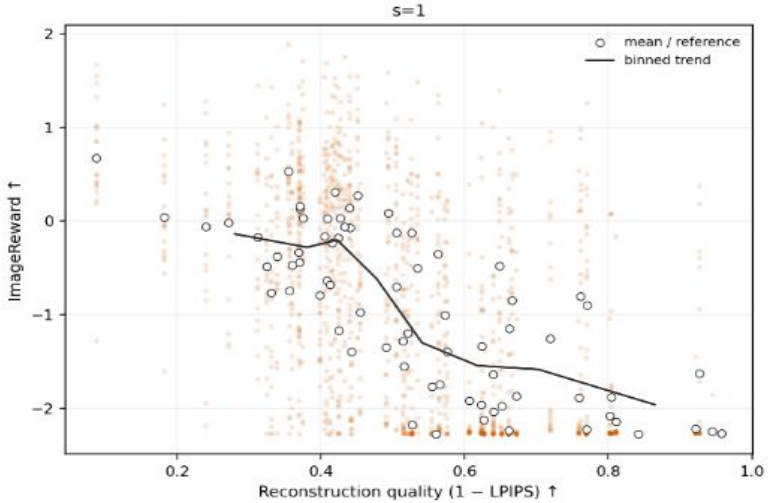

s=0
s=1
CLIP-Image ↑
CLIP-Text ↑
75.30
75.20
75.10
75.00
74.90
74.80
74.70
17.95
18.00
18.05
18.10
18.15
18.20
18.25
71.45
71.40
71.35
71.30
71.25
71.20
71.15
20.15
20.20
20.25
20.30
20.35
20.40
DDIM
Recap-LoRA
BLIP-LoRA
BLIP+style
BLIP-LoRA (adjacent z)
BLIP-LoRA (strided z)
BLIP+style LoRA (strided z)


s=0
s=1
CSD ↑
ImageReward ↑
0.650
0.649
0.648
0.647
0.646
0.645
0.644
0.643
-1.55
-1.54
-1.53
-1.52
0.614
0.613
0.612
0.611
0.610
-0.96
-0.95
-0.94
-0.93
-0.92
-0.91
-0.90
DDIM
Recap-LoRA
BLIP-LoRA
BLIP+style
BLIP-LoRA (adjacent z)
BLIP-LoRA (strided z)
BLIP+style LoRA (strided z)

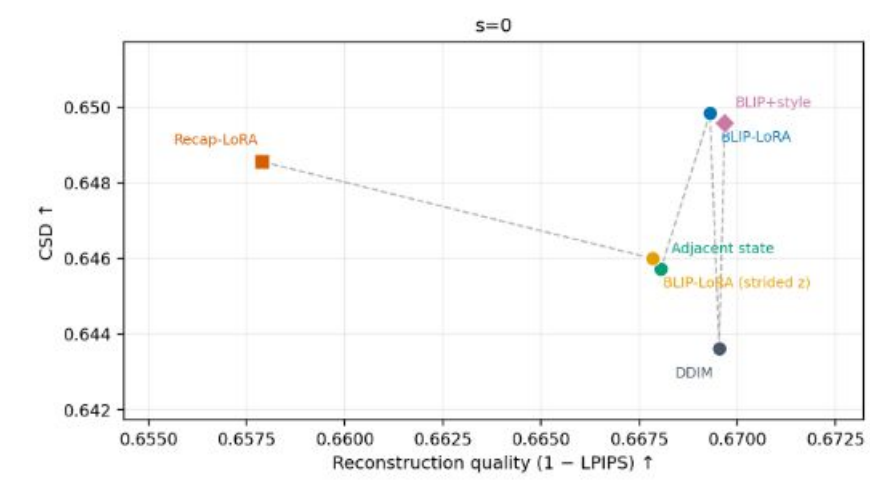

IP-Adapter
s=0
s=0.5
s=1
StyleAligned DDIM
StyleShot
InstantStyle
VisualStyle
CSGO
0.700
0.650
0.600
0.550
0.500
0.450
CSD ↑
-2.00
-1.50
-1.00
-0.50
0.00
ImageReward ↑
DDIM
Recap-LoRA
BLIP-LoRA
BLIP+style
BLIP-LoRA (adjacent z)
BLIP-LoRA (strided z)
BLIP+style LoRA (strided z)
II DDIM (empty)
II No-CFG LoRA (empty)
s=0
s=1

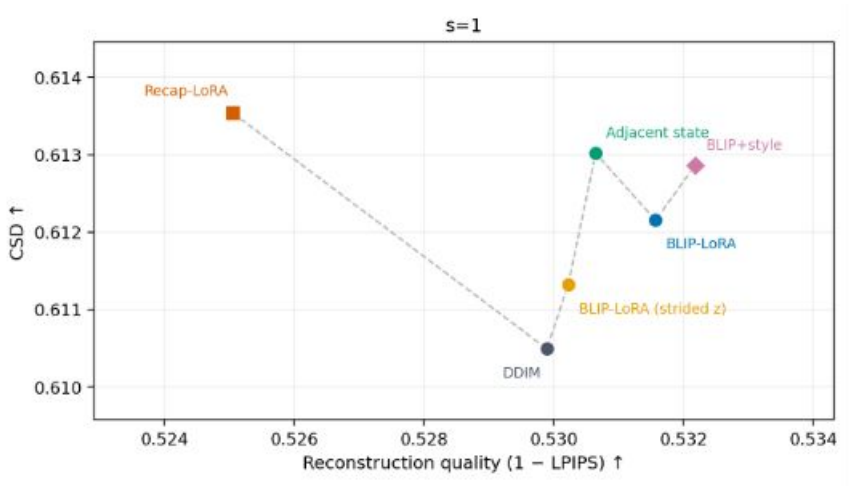


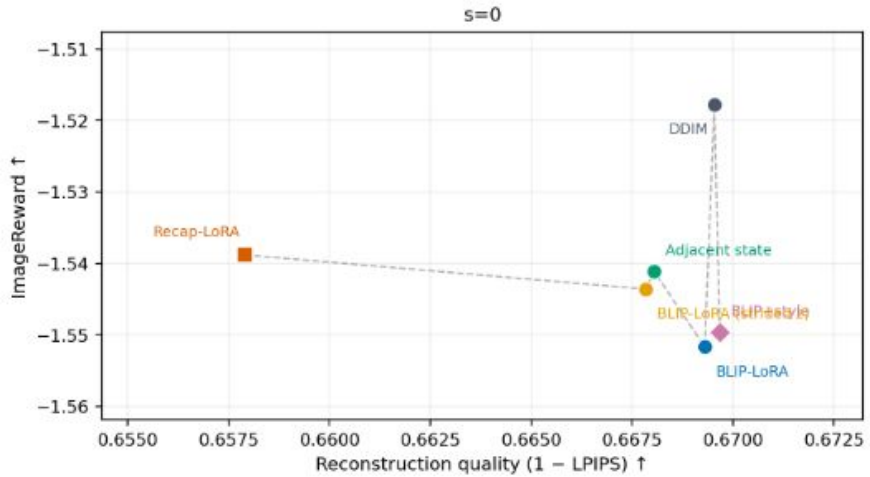


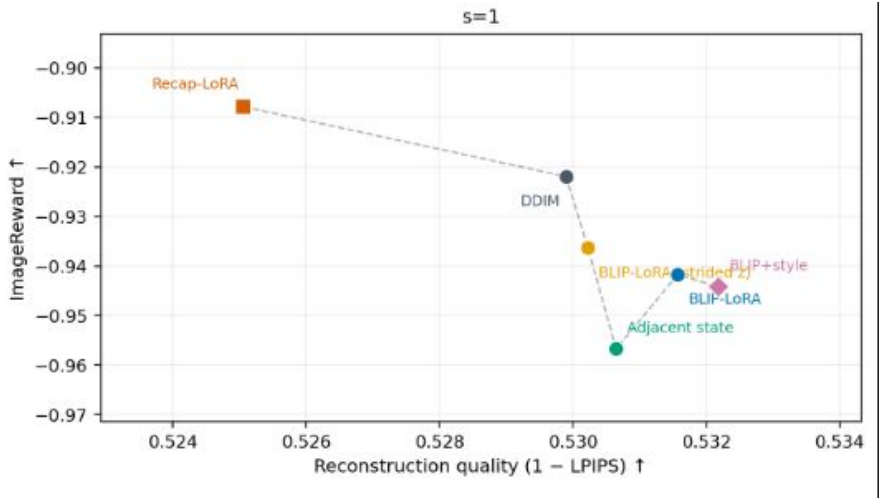


BLIP + styl LORA, analiza, rekonstruckaj vs transfer stylu

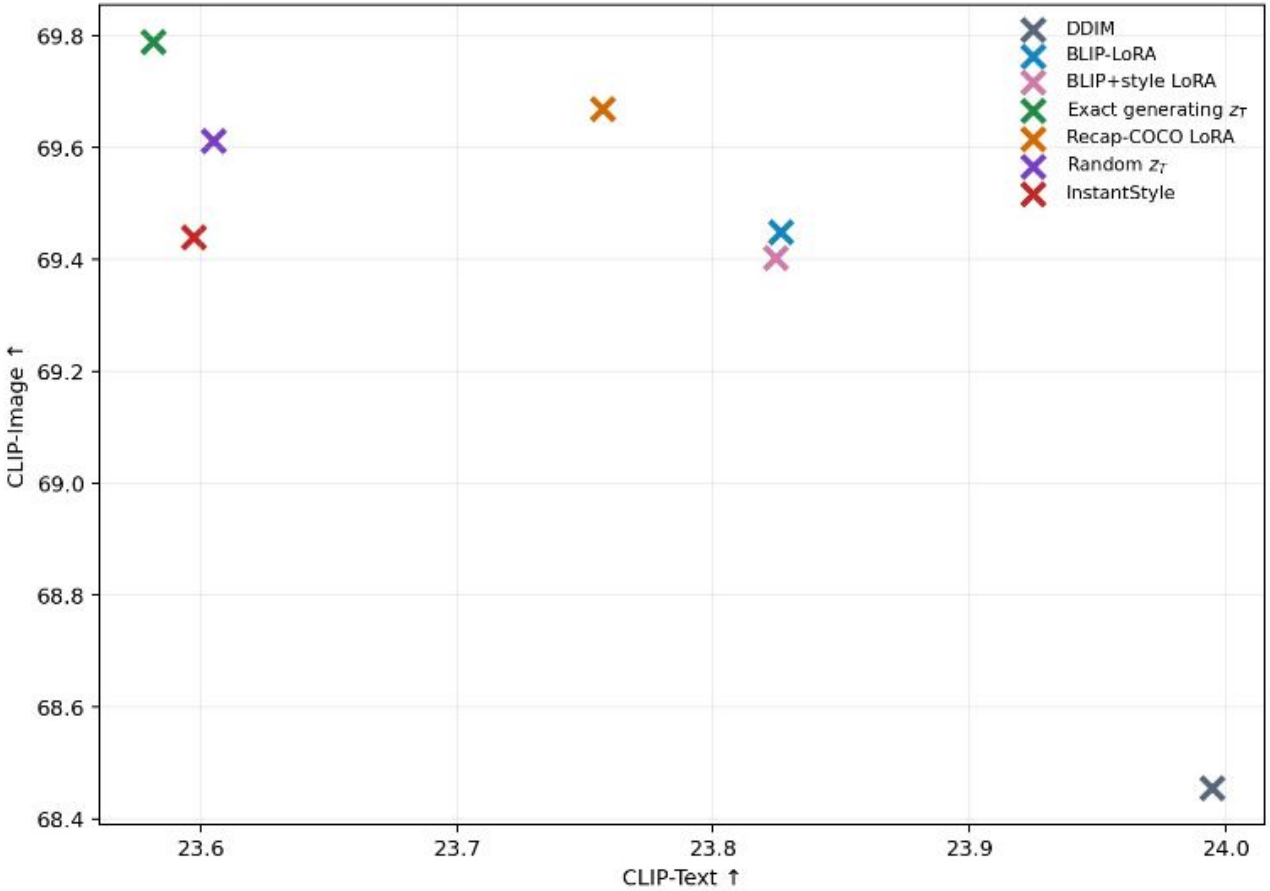

s=0
CSD ↑
Reconstruction quality (1 − LPIPS) ↑
Recap-LoRA
BLIP+style
BLIP-LoRA
Adjacent state
BLIP-LoRA (strided z)
DDIM

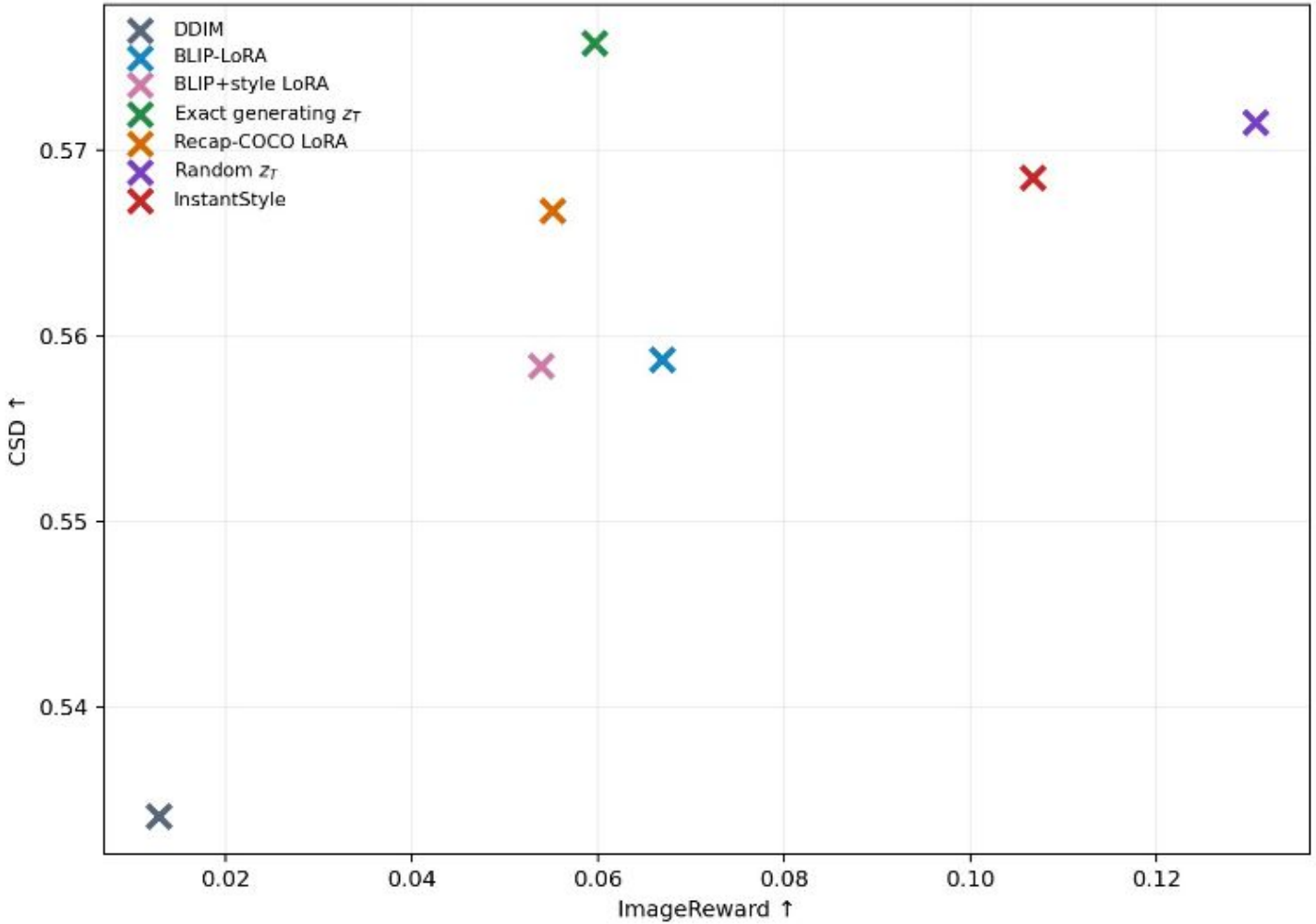

s=1
CSD ↑
Reconstruction quality (1 − LPIPS) ↑
Recap-LoRA
Adjacent state
BLIP+style
BLIP-LoRA
BLIP-LoRA (strided z)
DDIM

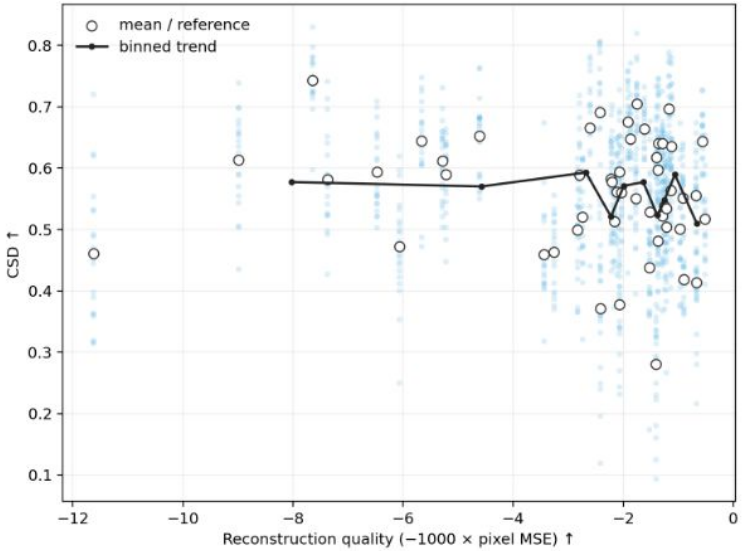

s=0
ImageReward ↑
Reconstruction quality (1 − LPIPS) ↑
DDIM
Recap-LoRA
Adjacent state
BLIP-LoRA
BLIP+style

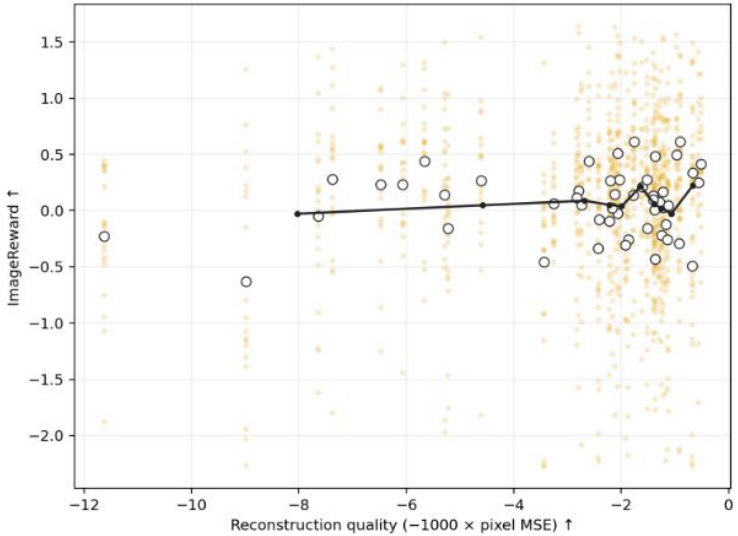

s=1
ImageReward ↑
Reconstruction quality (1 − LPIPS) ↑
Recap-LoRA
DDIM
BLIP+style
BLIP-LoRA
Adjacent state

Transfer stylu na wygenerowanych obrazakch (prompty z BLIP+styl z obrazków ze StyleBench)



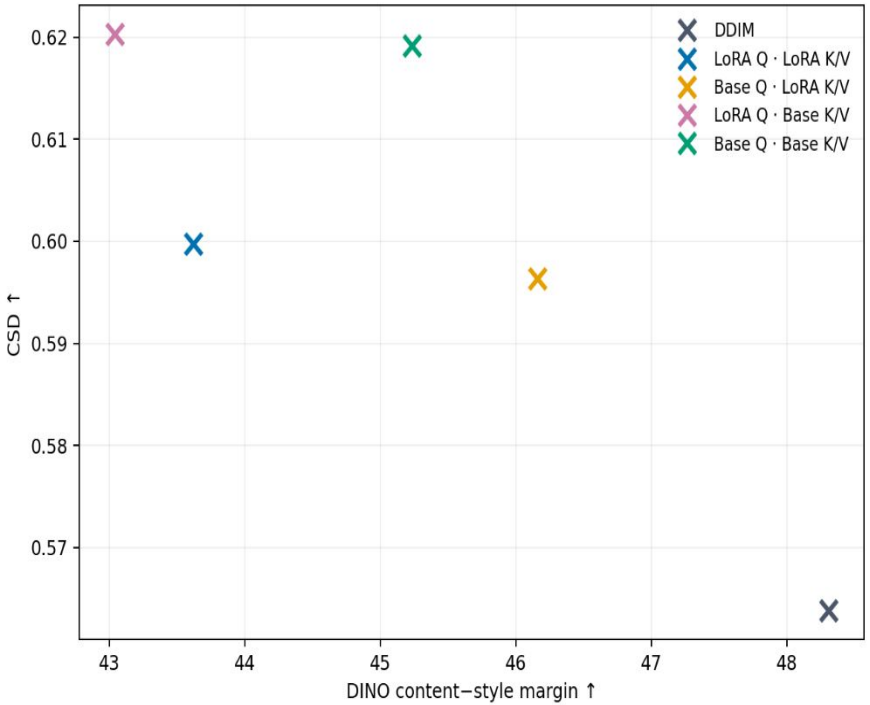

DDIM
BLIP-LoRA
BLIP+style LoRA
Exact generating z_T
Recap-COCO LoRA
Random z_T
InstantStyle
0.57
0.56
0.55
0.54
CSD ↑
0.02
0.04
0.06
0.08
0.10
0.12
ImageReward ↑


mean / reference
binned trend
CSD ↑
Reconstruction quality (−1000 × pixel MSE) ↑

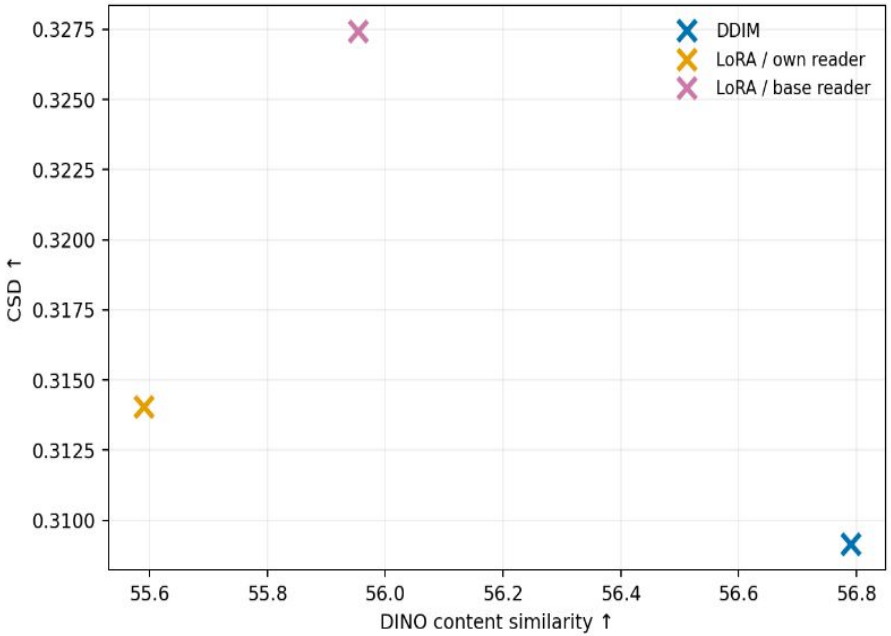

ImageReward ↑
Reconstruction quality (−1000 × pixel MSE) ↑

StyleID

wygenerowane



rzeczywiste



LoRA+DirectInv

- LoRA była puszczana i zapisywana w fp32 od początku - sam model był w bf16
- Do batcha wchodziły losowe trajektorie z losowych obrazów (nie z tego samego jak mówiłem)
- Dodany trening z modelem uruchamianym w fp32, batch size zwiększony do 128, trajektorie zapisywane w fp32, 1 epoka
- Dodatkowo trening z samplowaniem 40% przykładów treningowych z ostatnich 10% trajektorii

Training sampling / source	Inversion	α	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP target (edit) ↑
Paper original	DDIM	–	28.22	22.28	113.46	83.64	79.05	25.41	22.55
Paper original	DirectInv	–	24.29	**22.46**	**106.06**	**80.45**	79.68	25.41	22.62
Uniform	LoRA r8	4	**24.27** ±20.09	22.37 ± 4.49	110.17 ± 62.58	81.82 ± 60.23	**79.90** ±12.80	25.83 ± 3.73	22.85 ± 4.84
Uniform	LoRA r16	8							
Uniform	LoRA r32	16							
Final-tail	LoRA r8	4							
Final-tail	LoRA r16	8	24.44 ± 21.07	22.36 ± 4.47	110.71 ± 62.66	81.79 ± 60.00	79.88 ± 12.79	**25.84** ±3.74	22.87 ± 4.86
Final-tail	LoRA r32	16	24.45 ± 20.98	22.35 ± 4.44	110.79 ± 62.80	81.84 ± 60.04	79.88 ± 12.80	25.83 ± 3.72	**22.88** ±4.84

FP32 i Final-tail sampling

Editor	Training sampling / source	Inversion	α	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP target (edit) ↑
P2P	Paper original	DDIM	–	69.43	17.87	208.80	219.88	71.14	25.01	**22.44**
P2P	Paper original	DirectInv	–	**11.65**	**27.22**	**54.55**	**32.86**	**84.76**	25.02	22.10
P2P	Uniform	LoRA r8	4	14.12 ± 11.68	26.02 ± 5.36	73.25 ± 52.90	44.40 ± 47.79	83.50 ± 12.44	25.33 ± 3.79	22.31 ± 4.87
P2P	Uniform	LoRA r16	8	14.17 ± 11.78	26.01 ± 5.42	72.98 ± 50.27	43.99 ± 45.35	83.51 ± 12.24	25.29 ± 3.85	22.40 ± 4.86
P2P	Uniform	LoRA r32	16							
P2P	Final-tail	LoRA r8	4	14.27 ± 11.73	25.95 ± 5.30	72.48 ± 49.20	44.31 ± 46.18	83.55 ± 12.13	**25.35** ±3.77	22.37 ± 4.85
P2P	Final-tail	LoRA r16	8							
P2P	Final-tail	LoRA r32	16	14.37 ± 11.71	25.92 ± 5.38	74.82 ± 53.07	45.35 ± 48.56	83.34 ± 12.37	25.31 ± 3.83	22.33 ± 4.86
PnP	Paper original	DDIM	–	28.22	22.28	113.46	83.64	79.05	25.41	22.55
PnP	Paper original	DirectInv	–	24.29	**22.46**	**106.06**	**80.45**	79.68	25.41	22.62
PnP	Uniform	LoRA r8	4	**24.27** ±20.09	22.37 ± 4.49	110.17 ± 62.58	81.82 ± 60.23	79.90 ± 12.80	25.83 ± 3.73	22.85 ± 4.84
PnP	Uniform	LoRA r16	8	24.32 ± 20.11	22.37 ± 4.48	110.15 ± 62.53	81.75 ± 60.23	**79.90** ±12.81	25.79 ± 3.73	22.82 ± 4.86
PnP	Uniform	LoRA r32	16							
PnP	Final-tail	LoRA r8	4	24.54 ± 21.13	22.38 ± 4.45	110.40 ± 62.45	81.40 ± 59.74	79.88 ± 12.78	**25.85** ±3.74	22.88 ± 4.88
PnP	Final-tail	LoRA r16	8							
PnP	Final-tail	LoRA r32	16	24.43 ± 20.80	22.36 ± 4.46	110.82 ± 62.81	81.83 ± 60.02	79.88 ± 12.80	25.82 ± 3.71	**22.89** ±4.85
MasaCtrl	Paper original	DDIM	–	28.38	22.17	106.62	86.97	79.67	23.96	21.16
MasaCtrl	Paper original	DirectInv	–	**24.70**	**22.64**	**87.94**	**81.09**	**81.33**	24.38	21.35
MasaCtrl	Uniform	LoRA r8	4	28.58 ± 21.67	21.90 ± 4.93	99.01 ± 58.73	99.13 ± 84.36	80.49 ± 12.73	24.61 ± 3.65	**21.50** ±4.64
MasaCtrl	Uniform	LoRA r16	8	28.47 ± 21.63	21.91 ± 4.93	98.89 ± 58.83	98.90 ± 84.73	80.52 ± 12.72	24.61 ± 3.63	21.48 ± 4.67
MasaCtrl	Uniform	LoRA r32	16							
MasaCtrl	Final-tail	LoRA r8	4	28.38 ± 21.57	21.93 ± 4.93	98.72 ± 58.55	98.66 ± 84.98	80.54 ± 12.70	**24.64** ±3.65	21.46 ± 4.61
MasaCtrl	Final-tail	LoRA r16	8							
MasaCtrl	Final-tail	LoRA r32	16	28.33 ± 21.04	21.88 ± 4.87	98.96 ± 58.38	99.32 ± 85.64	80.49 ± 12.71	24.59 ± 3.63	21.49 ± 4.71
Pix2Pix-Zero	Paper original	DDIM	–	61.68	20.44	172.22	144.12	74.67	22.80	20.54
Pix2Pix-Zero	Paper original	DirectInv	–	**49.22**	**21.53**	**138.98**	**127.32**	77.05	23.31	**21.05**
Pix2Pix-Zero	Uniform	LoRA r8	4	52.60 ± 52.68	21.31 ± 5.45	144.57 ± 115.49	136.29 ± 157.55	77.14 ± 15.42	23.37 ± 4.38	20.96 ± 4.68
Pix2Pix-Zero	Uniform	LoRA r16	8	51.63 ± 50.83	21.25 ± 5.48	147.07 ± 117.89	137.65 ± 153.16	76.94 ± 15.62	23.39 ± 4.35	20.95 ± 4.64
Pix2Pix-Zero	Uniform	LoRA r32	16							
Pix2Pix-Zero	Final-tail	LoRA r8	4	51.67 ± 51.06	21.33 ± 5.43	144.87 ± 117.21	135.70 ± 153.36	**77.15** ±15.58	**23.56** ±4.29	21.01 ± 4.67
Pix2Pix-Zero	Final-tail	LoRA r16	8							
Pix2Pix-Zero	Final-tail	LoRA r32	16	52.21 ± 52.25	21.19 ± 5.50	147.18 ± 118.65	141.80 ± 163.40	76.91 ± 15.77	23.32 ± 4.31	20.96 ± 4.79

FLUX

Metoda	Normality		Image Reconstruction				CLIP Text Alignment		
	Corr. ↓	KL Div. $\times 10^2$ ↓	MAE ↓	LPIPS ↓	PSNR ↑	SSIM ↑	Source ↓	Target ↑	Directional ↑
guidance=1									
Gaussian	—	—	—	—	—	—	31.934 ± 12.327	81.292 ± 8.528	35.143 ± 6.413
Euler	0.13795	0.3796	0.08970	0.50005	18.97	0.63897	33.949 ± 12.531	78.456 ± 10.270	33.486 ± 6.886
Euler + nudging 1.5	0.28697	6.1737	0.03841	0.17837	26.17	0.84902	63.451 ± 12.967	47.924 ± 13.617	15.849 ± 8.525
LoRA	**0.13650**	**0.0397**	**0.01733**	**0.07333**	**32.43**	**0.95713**	**32.033** ± 12.287	**80.378** ± 8.662	**34.845** ± 6.396
guidance=3.5									
Gaussian	—	—	—	—	—	—	27.774 ± 11.887	82.984 ± 7.812	38.260 ± 5.748
Euler	0.18743	3.9522	0.21394	0.72322	11.20	0.43764	**33.071** ± 12.211	**73.917** ± 12.687	**32.813** ± 7.615
Euler + nudging 1.5	0.37150	3.2313	**0.06833**	**0.15837**	**22.93**	**0.80523**	72.051 ± 15.692	40.391 ± 16.287	9.840 ± 10.044
LoRA	**0.14628**	**0.8322**	0.26107	0.75262	9.47	0.39155	33.718 ± 11.329	67.161 ± 15.899	30.308 ± 8.064

Inversion g= 1, edit, generation 3.5

Metoda	Normality		Image Reconstruction				CLIP Text Alignment		
	Corr. ↓	KL Div. $\times 10^2$ ↓	MAE ↓	LPIPS ↓	PSNR ↑	SSIM ↑	Source ↓	Target ↑	Directional ↑
Gaussian	—	—	—	—	—	—	27.774 ± 11.887	82.984 ± 7.812	38.260 ± 5.748
Euler	0.17916	4.2695	0.06236	0.17645	20.85	**0.81535**	31.463 ± 12.809	78.916 ± 10.609	35.252 ± 6.979
Euler + nudging 1.5	0.36462	3.2371	0.09074	**0.15229**	19.56	0.78944	78.477 ± 12.430	36.375 ± 15.258	6.636 ± 8.530
LoRA	**0.14830**	**1.4746**	**0.06115**	0.17746	**20.97**	0.80636	**29.028** ± 12.987	**80.415** ± 9.879	**36.655** ± 6.554

PIE-Bench FLUX

Metoda	**Structure** ($\times 10^3$) ↓	**PSNR (dB)** ↑	**LPIPS** ($\times 10^3$) ↓	**MSE** ($\times 10^4$) ↓	**SSIM** ($\times 10^2$) ↑	**CLIP Whole** ↑	**CLIP Edited** ↑
Euler	71.20	17.28	245.79	281.38	69.99	23.68	21.44
Nudging ×1.5	**20.14**	**23.55**	**67.29**	**101.47**	**84.94**	23.20	20.38
LoRA $r = 16$, $\alpha = 8$	60.03	18.59	197.14	257.52	75.98	**24.33**	**21.73**

	Rekonstrukcja całego obrazu					**Tło po edycji**				**CLIP edycji**	
Metoda	PSNR ↑ dB	LPIPS ↓ ×1000	MSE ↓ ×10000	SSIM ↑ ×100	MAE ↓	PSNR ↑ dB	LPIPS ↓ ×1000	MSE ↓ ×10000	SSIM ↑ ×100	Cały obraz ↑	W masce ↑
Euler	16,88	329,96	284,88	60,14	0,11741	17,28	245,79	281,38	69,99	23,68	21,44
LoRA	20,95	194,15	180,07	75,42	0,07561	18,59	197,14	257,52	75,98	**24,33**	**21,73**
Nudging ×1,5	**22,10**	**82,27**	**115,19**	**79,11**	**0,06248**	**23,55**	**67,29**	**101,47**	**84,94**	23,20	20,38

PIE-Bench metrics split by real/generated images

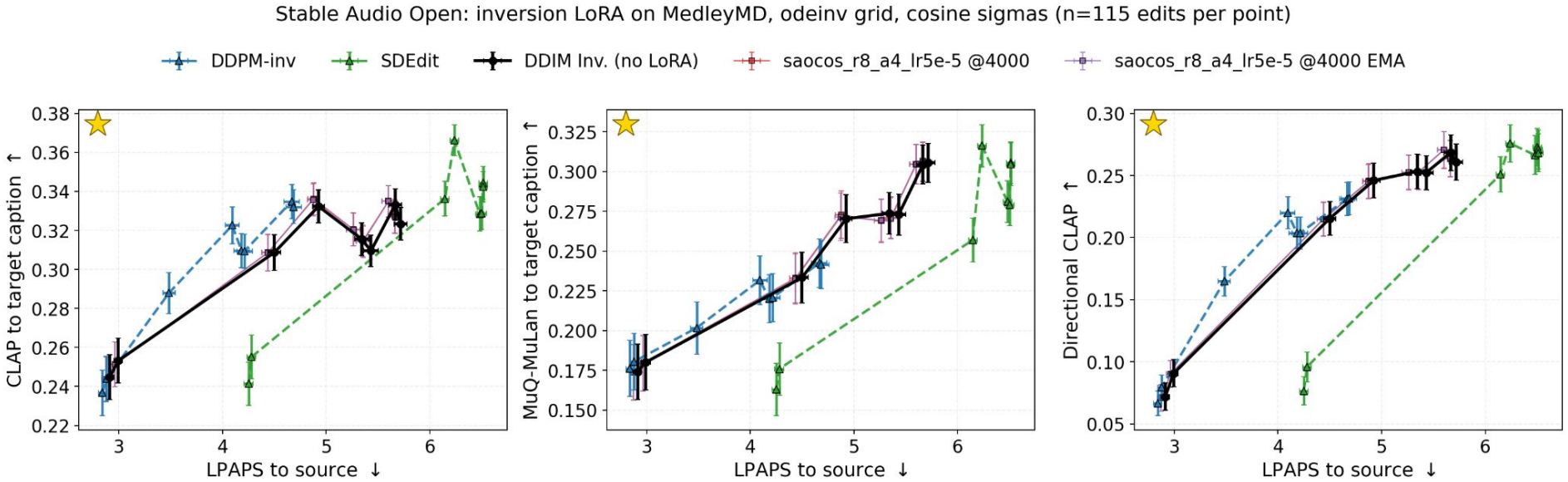
Editor	Training	Source	Structure↓	PSNR↑	LPIPS↓	MSE↓	SSIM↑	CLIP src↑	CLIP tgt↑	CLIP edit↑
P2P	Uniform, r16	Natural	**13.89 ± 12.23**	**26.28 ± 4.63**	**69.86 ± 49.21**	**37.96 ± 38.33**	**84.51 ± 10.96**	25.73 ± 3.10	24.59 ± 3.81	22.00 ± 4.89
P2P	Uniform, r16	Artificial	14.73 ± 11.80	24.82 ± 5.22	82.74 ± 53.65	54.64 ± 50.28	81.55 ± 13.09	**26.60 ± 3.31**	**25.99 ± 3.84**	**22.48 ± 5.13**
PnP	Uniform, r16	Natural	**22.85 ± 19.65**	**22.64 ± 3.34**	**106.61 ± 58.46**	**69.64 ± 44.71**	**81.28 ± 11.36**	25.73 ± 3.10	25.20 ± 3.54	22.60 ± 4.75
PnP	Uniform, r16	Artificial	26.29 ± 21.89	21.32 ± 4.05	119.01 ± 64.91	99.81 ± 66.82	77.66 ± 13.21	**26.60 ± 3.31**	**26.58 ± 3.77**	**22.89 ± 5.24**
MasaCtrl	Uniform, r16	Natural	**27.54 ± 20.13**	**22.10 ± 4.01**	**98.53 ± 56.72**	**87.94 ± 71.47**	**81.43 ± 11.14**	25.73 ± 3.10	24.23 ± 3.53	21.34 ± 4.74
MasaCtrl	Uniform, r16	Artificial	29.31 ± 22.55	21.21 ± 4.74	103.85 ± 61.96	116.09 ± 98.73	78.69 ± 13.55	**26.60 ± 3.31**	**25.31 ± 3.52**	**21.60 ± 4.92**
Pix2Pix-Zero	Uniform, r16	Natural	59.42 ± 57.60	**21.01 ± 5.17**	160.04 ± 126.33	144.01 ± 166.42	**76.56 ± 15.09**	25.73 ± 3.10	23.04 ± 4.21	**20.84 ± 4.66**
Pix2Pix-Zero	Uniform, r16	Artificial	**48.60 ± 44.24**	20.99 ± 5.33	**143.40 ± 115.56**	**140.69 ± 149.53**	75.99 ± 16.09	**26.60 ± 3.31**	**23.61 ± 4.51**	20.59 ± 4.80
P2P	Final-tail, r8	Natural	**13.86 ± 12.09**	**26.32 ± 4.55**	**67.71 ± 44.58**	**36.99 ± 36.80**	**84.65 ± 10.65**	25.73 ± 3.10	24.68 ± 3.81	21.99 ± 4.88
P2P	Final-tail, r8	Artificial	14.88 ± 11.68	24.72 ± 5.26	83.60 ± 54.77	56.23 ± 52.29	81.49 ± 13.08	**26.60 ± 3.31**	**26.05 ± 3.66**	**22.48 ± 5.06**
PnP	Final-tail, r8	Natural	**23.06 ± 20.36**	**22.66 ± 3.35**	**106.73 ± 58.34**	**69.37 ± 44.49**	**81.28 ± 11.32**	25.73 ± 3.10	25.27 ± 3.54	22.65 ± 4.76
PnP	Final-tail, r8	Artificial	26.42 ± 22.24	21.32 ± 4.02	119.65 ± 64.70	99.56 ± 66.24	77.61 ± 13.16	**26.60 ± 3.31**	**26.63 ± 3.80**	**23.01 ± 5.26**
MasaCtrl	Final-tail, r8	Natural	**27.33 ± 19.96**	**22.12 ± 3.97**	**97.98 ± 56.68**	**87.23 ± 71.13**	**81.47 ± 11.18**	25.73 ± 3.10	24.15 ± 3.55	21.31 ± 4.65
MasaCtrl	Final-tail, r8	Artificial	29.54 ± 23.11	21.21 ± 4.75	104.01 ± 61.37	116.35 ± 99.17	78.69 ± 13.48	**26.60 ± 3.31**	**25.46 ± 3.48**	**21.59 ± 4.86**
Pix2Pix-Zero	Final-tail, r8	Natural	58.99 ± 57.01	**21.16 ± 5.17**	157.46 ± 126.62	**139.37 ± 159.90**	**76.79 ± 15.22**	25.73 ± 3.10	23.16 ± 4.28	**20.88 ± 4.66**
Pix2Pix-Zero	Final-tail, r8	Artificial	**48.56 ± 45.15**	21.07 ± 5.40	**140.85 ± 112.27**	139.68 ± 153.13	76.29 ± 15.71	**26.60 ± 3.31**	**23.87 ± 4.25**	20.76 ± 4.86
P2P	Final-tail, r32	Natural	**13.91 ± 11.88**	**26.30 ± 4.65**	**69.71 ± 49.65**	**37.47 ± 36.07**	**84.44 ± 10.96**	25.73 ± 3.10	24.67 ± 3.90	21.91 ± 5.02
P2P	Final-tail, r32	Artificial	14.99 ± 11.96	24.70 ± 5.39	86.45 ± 59.13	58.27 ± 57.95	81.24 ± 13.36	**26.60 ± 3.31**	**25.97 ± 3.76**	**22.48 ± 5.03**
PnP	Final-tail, r32	Natural	**23.02 ± 20.35**	**22.63 ± 3.34**	**106.97 ± 58.66**	**69.73 ± 44.76**	**81.27 ± 11.34**	25.73 ± 3.10	25.28 ± 3.56	22.64 ± 4.77
PnP	Final-tail, r32	Artificial	26.24 ± 22.02	21.29 ± 4.01	120.27 ± 65.02	99.99 ± 66.31	77.60 ± 13.17	**26.60 ± 3.31**	**26.58 ± 3.68**	**23.00 ± 5.17**
MasaCtrl	Final-tail, r32	Natural	**27.30 ± 19.59**	**22.12 ± 3.98**	**97.85 ± 56.53**	**87.44 ± 71.59**	**81.45 ± 11.20**	25.73 ± 3.10	24.20 ± 3.49	21.33 ± 4.77
MasaCtrl	Final-tail, r32	Artificial	29.14 ± 21.80	21.15 ± 4.70	104.38 ± 60.80	116.96 ± 99.74	78.64 ± 13.48	**26.60 ± 3.31**	**25.35 ± 3.48**	**21.63 ± 4.95**
Pix2Pix-Zero	Final-tail, r32	Natural	59.92 ± 58.13	20.93 ± 5.24	161.11 ± 129.50	149.39 ± 175.89	**76.39 ± 15.28**	25.73 ± 3.10	22.91 ± 4.30	**20.84 ± 4.84**
Pix2Pix-Zero	Final-tail, r32	Artificial	**48.81 ± 47.92**	**21.10 ± 5.42**	**140.89 ± 113.28**	**141.71 ± 161.20**	76.27 ± 16.27	**26.60 ± 3.31**	**23.63 ± 4.38**	20.65 ± 4.95

Wyniki branch pair i CFG 2.5

Editor	Training / reference	Inversion	α	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP target (edit) ↑
P2P	Paper original	DDIM	–	69.43	17.87	208.80	219.88	71.14	25.01	22.44
P2P	Paper original	DirectInv	–	**11.65**	**27.22**	**54.55**	**32.86**	**84.76**	25.02	22.10
P2P	Best uniform CFG1	LoRA r8	4	14.12 ± 11.68	26.02 ± 5.36	73.25 ± 52.90	44.40 ± 47.79	83.50 ± 12.44	25.33 ± 3.79	22.31 ± 4.87
P2P	Best final-tail CFG1	LoRA r8	4	14.27 ± 11.73	25.95 ± 5.30	72.48 ± 49.20	44.31 ± 46.18	83.55 ± 12.13	**25.35 ± 3.77**	22.37 ± 4.85
P2P	Branch-pair CFG1	LoRA+DirectInv r8	4	199.60 ± 145.24	15.63 ± 6.33	374.03 ± 224.61	501.94 ± 419.94	62.97 ± 21.21	15.95 ± 5.05	16.15 ± 4.56
P2P	Branch-pair CFG1	LoRA+DirectInv r16	8	178.54 ± 121.54	15.71 ± 6.57	357.96 ± 211.65	502.96 ± 426.17	59.61 ± 22.91	16.61 ± 5.03	16.55 ± 4.55
P2P	CFG2.5 uniform	LoRA r8	4	71.55 ± 41.14	17.91 ± 3.97	211.94 ± 108.19	217.15 ± 143.27	71.80 ± 15.86	25.30 ± 3.68	**22.52 ± 4.51**
P2P	CFG2.5 uniform	LoRA r16	8	71.84 ± 41.54	17.90 ± 3.97	212.48 ± 108.77	218.15 ± 145.37	71.81 ± 15.90	25.26 ± 3.67	22.45 ± 4.53
P2P	CFG2.5 uniform	LoRA+DirectInv r8	4	15.93 ± 14.92	25.70 ± 5.51	81.20 ± 58.56	48.01 ± 50.79	82.78 ± 12.69	25.11 ± 3.88	22.23 ± 4.90
P2P	CFG2.5 uniform	LoRA+DirectInv r16	8	16.01 ± 15.55	25.57 ± 5.27	81.83 ± 57.95	48.72 ± 52.43	82.79 ± 12.54	25.22 ± 3.96	22.24 ± 4.97
P2P	CFG2.5 uniform	LoRA+DirectInv r32	16	16.21 ± 15.69	25.60 ± 5.47	82.51 ± 59.85	49.53 ± 55.11	82.66 ± 12.79	25.21 ± 3.95	22.29 ± 4.90
PnP	Paper original	DDIM	–	28.22	22.28	113.46	83.64	79.05	25.41	22.55
PnP	Paper original	DirectInv	–	24.29	22.46	106.06	80.45	79.68	25.41	22.62
PnP	Best uniform CFG1	LoRA r8	4	24.27 ± 20.09	22.37 ± 4.49	110.17 ± 62.58	81.82 ± 60.23	79.90 ± 12.80	**25.83 ± 3.73**	22.85 ± 4.84
PnP	Best final-tail CFG1	LoRA r32	16	24.43 ± 20.80	22.36 ± 4.46	110.82 ± 62.81	81.83 ± 60.02	79.88 ± 12.80	25.82 ± 3.71	**22.89 ± 4.85**
PnP	Branch-pair CFG1	LoRA+DirectInv r8	4	271.82 ± 160.47	12.07 ± 3.72	475.18 ± 212.15	810.72 ± 487.99	55.82 ± 20.42	12.26 ± 3.38	15.28 ± 4.36
PnP	Branch-pair CFG1	LoRA+DirectInv r32	16	272.46 ± 158.92	14.06 ± 3.98	426.96 ± 208.83	523.24 ± 347.38	66.40 ± 18.75	12.78 ± 3.69	15.71 ± 4.36
PnP	CFG2.5 uniform	LoRA r8	4	27.29 ± 25.88	**22.60 ± 4.61**	109.23 ± 62.19	**78.07 ± 56.51**	80.14 ± 12.78	25.60 ± 3.67	22.59 ± 4.83
PnP	CFG2.5 uniform	LoRA r16	8	27.24 ± 25.73	22.60 ± 4.60	109.61 ± 62.71	78.27 ± 57.06	80.13 ± 12.82	25.53 ± 3.69	22.54 ± 4.84
PnP	CFG2.5 uniform	LoRA+DirectInv r8	4	23.77 ± 20.03	22.59 ± 4.52	104.66 ± 60.25	78.24 ± 57.77	**80.52 ± 12.71**	25.62 ± 3.74	22.65 ± 4.88
PnP	CFG2.5 uniform	LoRA+DirectInv r16	8	23.79 ± 20.27	22.57 ± 4.44	104.43 ± 60.21	78.27 ± 57.91	80.47 ± 12.73	25.61 ± 3.73	22.62 ± 4.87
PnP	CFG2.5 uniform	LoRA+DirectInv r32	16	**23.71 ± 20.25**	22.56 ± 4.45	**104.37 ± 60.08**	78.48 ± 58.08	80.45 ± 12.71	25.62 ± 3.69	22.64 ± 4.85
MasaCtrl	Paper original	DDIM	–	28.38	22.17	106.62	86.97	79.67	23.96	21.16
MasaCtrl	Paper original	DirectInv	–	**24.70**	**22.64**	**87.94**	**81.09**	**81.33**	24.38	21.35
MasaCtrl	Best uniform CFG1	LoRA r16	8	28.47 ± 21.63	21.91 ± 4.93	98.89 ± 58.83	98.90 ± 84.73	80.52 ± 12.72	**24.61 ± 3.63**	21.48 ± 4.67
MasaCtrl	Best final-tail CFG1	LoRA r32	16	28.33 ± 21.04	21.88 ± 4.87	98.96 ± 58.38	99.32 ± 85.64	80.49 ± 12.71	24.59 ± 3.63	**21.49 ± 4.71**
MasaCtrl	Branch-pair CFG1	LoRA+DirectInv r16	8	212.08 ± 115.07	13.37 ± 4.03	400.75 ± 185.79	603.56 ± 365.06	55.95 ± 21.38	14.36 ± 3.69	16.12 ± 4.27
MasaCtrl	CFG2.5 uniform	LoRA r8	4	55.94 ± 37.74	20.30 ± 4.42	171.62 ± 90.65	131.29 ± 99.02	75.14 ± 15.10	21.08 ± 4.50	19.34 ± 4.59
MasaCtrl	CFG2.5 uniform	LoRA r16	8	57.89 ± 39.41	20.31 ± 4.44	172.45 ± 91.59	131.85 ± 100.57	75.12 ± 15.23	21.07 ± 4.44	19.30 ± 4.59
MasaCtrl	CFG2.5 uniform	LoRA+DirectInv r8	4	35.52 ± 24.83	21.04 ± 4.97	114.72 ± 67.61	121.57 ± 105.54	78.80 ± 13.51	24.27 ± 3.68	21.32 ± 4.58
MasaCtrl	CFG2.5 uniform	LoRA+DirectInv r16	8	35.84 ± 25.05	21.01 ± 4.97	115.20 ± 67.94	122.34 ± 107.22	78.76 ± 13.52	24.24 ± 3.64	21.29 ± 4.65
MasaCtrl	CFG2.5 uniform	LoRA+DirectInv r32	16	35.86 ± 25.45	20.99 ± 4.96	115.41 ± 67.71	122.70 ± 107.58	78.72 ± 13.51	24.20 ± 3.66	21.25 ± 4.64
Pix2Pix-Zero	Paper original	DDIM	–	61.68	20.44	172.22	144.12	74.67	22.80	20.54
Pix2Pix-Zero	Paper original	DirectInv	–	**49.22**	**21.53**	**138.98**	**127.32**	77.05	23.31	**21.05**
Pix2Pix-Zero	Best uniform CFG1	LoRA r16	8	51.63 ± 50.83	21.25 ± 5.48	147.07 ± 117.89	137.65 ± 153.16	76.94 ± 15.62	23.39 ± 4.35	20.95 ± 4.64
Pix2Pix-Zero	Best final-tail CFG1	LoRA r8	4	51.67 ± 51.06	21.33 ± 5.43	144.87 ± 117.21	135.70 ± 153.36	**77.15 ± 15.58**	**23.56 ± 4.29**	21.01 ± 4.67
Pix2Pix-Zero	CFG2.5 uniform	LoRA r8	4	65.88 ± 48.83	20.50 ± 4.86	182.65 ± 117.39	148.20 ± 157.13	75.02 ± 15.72	22.91 ± 4.55	20.65 ± 4.73
Pix2Pix-Zero	CFG2.5 uniform	LoRA+DirectInv r8	4	55.93 ± 55.15	20.93 ± 5.53	152.91 ± 121.24	151.32 ± 175.20	76.47 ± 15.87	23.12 ± 4.37	20.85 ± 4.83
Pix2Pix-Zero	CFG2.5 uniform	LoRA+DirectInv r16	8	56.23 ± 55.10	20.81 ± 5.45	153.89 ± 121.81	153.21 ± 172.14	76.46 ± 15.89	23.17 ± 4.49	20.85 ± 4.76
Pix2Pix-Zero	CFG2.5 uniform	LoRA+DirectInv r32	16	58.23 ± 60.19	20.89 ± 5.55	154.12 ± 123.21	152.52 ± 171.04	76.38 ± 15.98	23.02 ± 4.61	20.75 ± 4.90

Pnp - sd1.5
MasaCtrl, p2p, pix2pix-zero - sd1.4,

Editor	Training / reference	Inversion	α	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP target (edit) ↑
P2P	SD1.5 reproduced	DDIM	–	70.50 ± 38.97	17.84 ± 3.93	210.15 ± 107.16	220.28 ± 145.34	71.56 ± 15.89	25.19 ± 3.68	22.39 ± 4.48
P2P	SD1.5 reproduced	DirectInv	–	31.35 ± 25.29	22.57 ± 5.10	125.51 ± 82.42	91.27 ± 85.39	78.54 ± 14.04	25.07 ± 3.79	22.22 ± 4.79
P2P	Best uniform CFG1	LoRA+DirectInv r8	4	**14.12 ± 11.68**	**26.02 ± 5.36**	73.25 ± 52.90	44.40 ± 47.79	83.50 ± 12.44	25.33 ± 3.79	22.31 ± 4.87
P2P	Best final-tail CFG1	LoRA+DirectInv r8	4	14.27 ± 11.73	25.95 ± 5.30	**72.48 ± 49.20**	**44.31 ± 46.18**	**83.55 ± 12.13**	**25.35 ± 3.77**	22.37 ± 4.85
P2P	Branch-pair CFG1	LoRA+DirectInv r8	4	199.60 ± 145.24	15.63 ± 6.33	374.03 ± 224.61	501.94 ± 419.94	62.97 ± 21.21	15.95 ± 5.05	16.15 ± 4.56
P2P	Branch-pair CFG1	LoRA+DirectInv r16	8	178.54 ± 121.54	15.71 ± 6.57	357.96 ± 211.65	502.96 ± 426.17	59.61 ± 22.91	16.61 ± 5.03	16.55 ± 4.55
P2P	CFG2.5 uniform	LoRA r8	4	71.55 ± 41.14	17.91 ± 3.97	211.94 ± 108.19	217.15 ± 143.27	71.80 ± 15.86	25.30 ± 3.68	**22.52 ± 4.51**
P2P	CFG2.5 uniform	LoRA r16	8	71.84 ± 41.54	17.90 ± 3.97	212.48 ± 108.77	218.15 ± 145.37	71.81 ± 15.90	25.26 ± 3.67	22.45 ± 4.53
P2P	CFG2.5 uniform	LoRA+DirectInv r8	4	15.93 ± 14.92	25.70 ± 5.51	81.20 ± 58.56	48.01 ± 50.79	82.78 ± 12.69	25.11 ± 3.88	22.23 ± 4.90
P2P	CFG2.5 uniform	LoRA+DirectInv r16	8	16.01 ± 15.55	25.57 ± 5.27	81.83 ± 57.95	48.72 ± 52.43	82.79 ± 12.54	25.22 ± 3.96	22.24 ± 4.97
P2P	CFG2.5 uniform	LoRA+DirectInv r32	16	16.21 ± 15.69	25.60 ± 5.47	82.51 ± 59.85	49.53 ± 55.11	82.66 ± 12.79	25.21 ± 3.95	22.29 ± 4.90
PnP	SD1.5 reproduced	DDIM	–	27.32 ± 24.59	22.31 ± 4.50	112.76 ± 64.87	82.98 ± 60.97	79.58 ± 12.90	25.59 ± 3.73	22.57 ± 4.86
PnP	SD1.5 reproduced	DirectInv	–	**23.33 ± 20.19**	22.46 ± 4.34	105.51 ± 59.90	79.97 ± 59.13	80.22 ± 12.76	25.63 ± 3.78	22.67 ± 4.89
PnP	Best uniform CFG1	LoRA+DirectInv r8	4	24.27 ± 20.09	22.37 ± 4.49	110.17 ± 62.58	81.82 ± 60.23	79.90 ± 12.80	**25.83 ± 3.73**	22.85 ± 4.84
PnP	Best final-tail CFG1	LoRA+DirectInv r32	16	24.43 ± 20.80	22.36 ± 4.46	110.82 ± 62.81	81.83 ± 60.02	79.88 ± 12.80	25.82 ± 3.71	**22.89 ± 4.85**
PnP	Branch-pair CFG1	LoRA+DirectInv r8	4	271.82 ± 160.47	12.07 ± 3.72	475.18 ± 212.15	810.72 ± 487.99	55.82 ± 20.42	12.26 ± 3.38	15.28 ± 4.36
PnP	Branch-pair CFG1	LoRA+DirectInv r32	16	272.46 ± 158.92	14.06 ± 3.98	426.96 ± 208.83	523.24 ± 347.38	66.40 ± 18.75	12.78 ± 3.69	15.71 ± 4.36
PnP	CFG2.5 uniform	LoRA r8	4	27.29 ± 25.88	**22.60 ± 4.61**	109.23 ± 62.19	**78.07 ± 56.51**	80.14 ± 12.78	25.60 ± 3.67	22.59 ± 4.83
PnP	CFG2.5 uniform	LoRA r16	8	27.24 ± 25.73	**22.60 ± 4.60**	109.61 ± 62.71	78.27 ± 57.06	80.13 ± 12.82	25.53 ± 3.69	22.54 ± 4.84
PnP	CFG2.5 uniform	LoRA+DirectInv r8	4	23.77 ± 20.03	22.59 ± 4.52	104.66 ± 60.25	78.24 ± 57.77	**80.52 ± 12.71**	25.62 ± 3.74	22.65 ± 4.88
PnP	CFG2.5 uniform	LoRA+DirectInv r16	8	23.79 ± 20.27	22.57 ± 4.44	104.43 ± 60.21	78.27 ± 57.91	80.47 ± 12.73	25.61 ± 3.73	22.62 ± 4.87
PnP	CFG2.5 uniform	LoRA+DirectInv r32	16	23.71 ± 20.25	22.56 ± 4.45	**104.37 ± 60.08**	78.48 ± 58.08	80.45 ± 12.71	25.62 ± 3.69	22.64 ± 4.85
MasaCtrl	SD1.5 reproduced	DDIM	–	27.68 ± 21.08	22.32 ± 4.52	105.51 ± 60.03	84.69 ± 66.87	80.41 ± 12.65	24.16 ± 3.86	21.20 ± 4.71
MasaCtrl	SD1.5 reproduced	DirectInv	–	**23.57 ± 19.02**	**22.82 ± 4.87**	**86.83 ± 51.98**	**79.19 ± 67.20**	**82.02 ± 12.22**	24.51 ± 3.66	21.47 ± 4.61
MasaCtrl	Best uniform CFG1	LoRA+DirectInv r16	8	28.47 ± 21.63	21.91 ± 4.93	98.89 ± 58.83	98.90 ± 84.73	80.52 ± 12.72	**24.61 ± 3.63**	21.48 ± 4.67
MasaCtrl	Best final-tail CFG1	LoRA+DirectInv r32	16	28.33 ± 21.04	21.88 ± 4.87	98.96 ± 58.38	99.32 ± 85.64	80.49 ± 12.71	24.59 ± 3.63	**21.49 ± 4.71**
MasaCtrl	Branch-pair CFG1	LoRA+DirectInv r16	8	212.08 ± 115.07	13.37 ± 4.03	400.75 ± 185.79	603.56 ± 365.06	55.95 ± 21.38	14.36 ± 3.69	16.12 ± 4.27
MasaCtrl	CFG2.5 uniform	LoRA r8	4	55.94 ± 37.74	20.30 ± 4.42	171.62 ± 90.65	131.29 ± 99.02	75.14 ± 15.10	21.08 ± 4.50	19.34 ± 4.59
MasaCtrl	CFG2.5 uniform	LoRA r16	8	57.89 ± 39.41	20.31 ± 4.44	172.45 ± 91.59	131.85 ± 100.57	75.12 ± 15.23	21.07 ± 4.44	19.30 ± 4.59
MasaCtrl	CFG2.5 uniform	LoRA+DirectInv r8	4	35.52 ± 24.83	21.04 ± 4.97	114.72 ± 67.61	121.57 ± 105.54	78.80 ± 13.51	24.27 ± 3.68	21.32 ± 4.58
MasaCtrl	CFG2.5 uniform	LoRA+DirectInv r16	8	35.84 ± 25.05	21.01 ± 4.97	115.20 ± 67.94	122.34 ± 107.22	78.76 ± 13.52	24.24 ± 3.64	21.29 ± 4.65
MasaCtrl	CFG2.5 uniform	LoRA+DirectInv r32	16	35.86 ± 25.45	20.99 ± 4.96	115.41 ± 67.71	122.70 ± 107.58	78.72 ± 13.51	24.20 ± 3.66	21.25 ± 4.64
Pix2Pix-Zero	SD1.5 reproduced	DDIM	–	–	–	–	–	–	–	–
Pix2Pix-Zero	SD1.5 reproduced	DirectInv	–	**49.99 ± 51.39**	**21.47 ± 5.49**	**142.46 ± 115.06**	**133.49 ± 154.49**	**77.49 ± 15.64**	23.53 ± 4.40	**21.09 ± 4.77**
Pix2Pix-Zero	Best uniform CFG1	LoRA+DirectInv r16	8	51.63 ± 50.83	21.25 ± 5.48	147.07 ± 117.89	137.65 ± 153.16	76.94 ± 15.62	23.39 ± 4.35	20.95 ± 4.64
Pix2Pix-Zero	Best final-tail CFG1	LoRA+DirectInv r8	4	51.67 ± 51.06	21.33 ± 5.43	144.87 ± 117.21	135.70 ± 153.36	77.15 ± 15.58	**23.56 ± 4.29**	21.01 ± 4.67
Pix2Pix-Zero	CFG2.5 uniform	LoRA r8	4	65.88 ± 48.83	20.50 ± 4.86	182.65 ± 117.39	148.20 ± 157.13	75.02 ± 15.72	22.91 ± 4.55	20.65 ± 4.73
Pix2Pix-Zero	CFG2.5 uniform	LoRA+DirectInv r8	4	55.93 ± 55.15	20.93 ± 5.53	152.91 ± 121.24	151.32 ± 175.20	76.47 ± 15.87	23.12 ± 4.37	20.85 ± 4.83
Pix2Pix-Zero	CFG2.5 uniform	LoRA+DirectInv r16	8	56.23 ± 55.10	20.81 ± 5.45	153.89 ± 121.81	153.21 ± 172.14	76.46 ± 15.89	23.17 ± 4.49	20.85 ± 4.76
Pix2Pix-Zero	CFG2.5 uniform	LoRA+DirectInv r32	16	58.23 ± 60.19	20.89 ± 5.55	154.12 ± 123.21	152.52 ± 171.04	76.38 ± 15.98	23.02 ± 4.61	20.75 ± 4.90


Stable Audio Open: inversion LoRA on MedleyMD, odeinv grid, cosine sigmas (n=115 edits per point)
DDPM-inv
SDEdit
DDIM Inv. (no LoRA)
saocos_r8_a4_lr5e-5 @4000
saocos_r8_a4_lr5e-5 @4000 EMA
CLAP to target caption ↑
MuQ-MuLan to target caption ↑
Directional CLAP ↑
LPAPS to source ↓

Cycle-loss results

$$\mathcal{L}_{\text{total}}(\phi) = \underbrace{\|\varepsilon_\phi(z_{i+1}, t_i, c) - \varepsilon_\theta(z_i, t_i, c)\|_2^2}_{\mathcal{L}_{\text{inv}}} + \lambda_{\text{cycle}} \underbrace{\left\| z_{i+1} - F_\theta^{\text{gen}}\left(F_\phi^{\text{inv}}(z_{i+1}, t_i, c), t_i, c\right)\right\|}_{\mathcal{L}_{\text{cycle}}}$$

Editor	Training / reference	Inversion	α	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP target (edit) ↑
P2P	SD1.5 reproduced	DDIM	–	70.50 ± 38.97	17.84 ± 3.93	210.15 ± 107.16	220.28 ± 145.34	71.56 ± 15.89	25.19 ± 3.68	22.39 ± 4.48
P2P	SD1.5 reproduced	DirectInv	–	31.35 ± 25.29	22.57 ± 5.10	125.51 ± 82.42	91.27 ± 85.39	78.54 ± 14.04	25.07 ± 3.79	22.22 ± 4.79
P2P	Best uniform CFG1	LoRA+DirectInv r8	4	14.12 ± 11.68	26.02 ± 5.36	73.25 ± 52.90	44.40 ± 47.79	83.50 ± 12.44	25.33 ± 3.79	22.31 ± 4.87
P2P	Best final-tail CFG1	LoRA+DirectInv r8	4	14.27 ± 11.73	25.95 ± 5.30	72.48 ± 49.20	44.31 ± 46.18	83.55 ± 12.13	25.35 ± 3.77	22.37 ± 4.85
P2P	Cycle loss ($\lambda = 1$)	LoRA r8	4	70.40 ± 38.78	17.82 ± 3.95	211.23 ± 106.60	221.46 ± 146.67	71.35 ± 15.81	25.37 ± 3.67	22.64 ± 4.50
P2P	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r8	4	**14.01 ± 11.38**	**26.03 ± 5.43**	**72.35 ± 49.47**	**43.83 ± 45.13**	**83.56 ± 12.16**	25.32 ± 3.83	22.31 ± 4.88
P2P	Cycle loss ($\lambda = 1$)	LoRA r16	8	70.57 ± 39.83	17.82 ± 3.94	211.10 ± 106.78	221.37 ± 146.41	71.34 ± 15.81	**25.42 ± 3.71**	**22.65 ± 4.57**
P2P	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r16	8	14.45 ± 11.85	25.92 ± 5.39	74.51 ± 52.35	45.37 ± 46.89	83.37 ± 12.32	25.27 ± 3.84	22.31 ± 4.84
P2P	Cycle loss ($\lambda = 1$)	LoRA r32	16	70.33 ± 39.36	17.83 ± 3.94	210.93 ± 106.55	220.31 ± 145.03	71.40 ± 15.81	25.37 ± 3.64	**22.65 ± 4.52**
P2P	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r32	16	14.04 ± 11.47	25.99 ± 5.40	72.48 ± 49.20	43.89 ± 44.49	83.53 ± 12.21	25.31 ± 3.85	22.34 ± 4.89
PnP	SD1.5 reproduced	DDIM	–	27.32 ± 24.59	22.31 ± 4.50	112.76 ± 64.87	82.98 ± 60.97	79.58 ± 12.90	25.59 ± 3.73	22.57 ± 4.86
PnP	SD1.5 reproduced	DirectInv	–	**23.33 ± 20.19**	**22.46 ± 4.34**	**105.51 ± 59.90**	**79.97 ± 59.13**	**80.22 ± 12.76**	25.63 ± 3.78	22.67 ± 4.89
PnP	Best uniform CFG1	LoRA+DirectInv r8	4	24.27 ± 20.09	22.37 ± 4.49	110.17 ± 62.58	81.82 ± 60.23	79.90 ± 12.80	**25.83 ± 3.73**	22.85 ± 4.84
PnP	Best final-tail CFG1	LoRA+DirectInv r32	16	24.43 ± 20.80	22.36 ± 4.46	110.82 ± 62.81	81.83 ± 60.02	79.88 ± 12.80	25.82 ± 3.71	**22.89 ± 4.85**
PnP	Cycle loss ($\lambda = 1$)	LoRA r8	4	27.17 ± 24.47	22.32 ± 4.53	113.87 ± 65.57	82.79 ± 59.84	79.50 ± 12.90	25.71 ± 3.77	22.77 ± 4.86
PnP	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r8	4	24.19 ± 19.84	22.38 ± 4.50	110.03 ± 62.50	81.64 ± 60.21	79.92 ± 12.80	25.82 ± 3.72	**22.89 ± 4.86**
PnP	Cycle loss ($\lambda = 1$)	LoRA r16	8	27.02 ± 24.39	22.30 ± 4.51	113.75 ± 64.70	82.78 ± 59.42	79.52 ± 12.83	25.72 ± 3.76	22.76 ± 4.86
PnP	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r16	8	24.38 ± 20.67	22.36 ± 4.48	110.28 ± 62.53	81.87 ± 60.30	79.88 ± 12.81	25.81 ± 3.74	22.83 ± 4.86
PnP	Cycle loss ($\lambda = 1$)	LoRA r32	16	26.94 ± 24.29	22.33 ± 4.48	113.26 ± 64.71	82.20 ± 59.26	79.58 ± 12.84	25.68 ± 3.77	22.76 ± 4.90
PnP	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r32	16	24.17 ± 20.07	22.38 ± 4.50	110.13 ± 62.39	81.72 ± 60.19	79.91 ± 12.80	25.78 ± 3.75	22.83 ± 4.86
MasaCtrl	SD1.5 reproduced	DDIM	–	27.68 ± 21.08	22.32 ± 4.52	105.51 ± 60.03	84.69 ± 66.87	80.41 ± 12.65	24.16 ± 3.86	21.20 ± 4.71
MasaCtrl	SD1.5 reproduced	DirectInv	–	**23.57 ± 19.02**	**22.82 ± 4.87**	**86.83 ± 51.98**	**79.19 ± 67.20**	**82.02 ± 12.22**	24.51 ± 3.66	21.47 ± 4.61
MasaCtrl	Best uniform CFG1	LoRA+DirectInv r16	8	28.47 ± 21.63	21.91 ± 4.93	98.89 ± 58.83	98.90 ± 84.73	80.52 ± 12.72	24.61 ± 3.63	21.48 ± 4.67
MasaCtrl	Best final-tail CFG1	LoRA+DirectInv r32	16	28.33 ± 21.04	21.88 ± 4.87	98.96 ± 58.38	99.32 ± 85.64	80.49 ± 12.71	24.59 ± 3.63	21.49 ± 4.71
MasaCtrl	Cycle loss ($\lambda = 1$)	LoRA r8	4	49.50 ± 30.08	19.96 ± 4.43	179.40 ± 91.98	141.84 ± 105.95	73.97 ± 15.32	20.62 ± 4.45	19.22 ± 4.66
MasaCtrl	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r8	4	28.60 ± 21.70	21.91 ± 4.93	98.73 ± 58.98	98.95 ± 85.02	80.54 ± 12.72	24.60 ± 3.65	**21.50 ± 4.62**
MasaCtrl	Cycle loss ($\lambda = 1$)	LoRA r16	8	49.39 ± 29.59	19.93 ± 4.37	180.12 ± 92.26	142.61 ± 106.72	73.89 ± 15.36	20.61 ± 4.55	19.22 ± 4.72
MasaCtrl	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r16	8	28.49 ± 21.61	21.89 ± 4.89	98.91 ± 58.48	98.95 ± 84.46	80.50 ± 12.73	**24.63 ± 3.64**	21.48 ± 4.67
MasaCtrl	Cycle loss ($\lambda = 1$)	LoRA r32	16	49.59 ± 29.73	19.94 ± 4.32	179.67 ± 91.96	141.88 ± 105.82	73.95 ± 15.36	20.55 ± 4.60	19.21 ± 4.68
MasaCtrl	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r32	16	28.48 ± 21.49	21.90 ± 4.88	98.65 ± 58.25	98.72 ± 84.36	80.53 ± 12.68	24.62 ± 3.61	21.49 ± 4.68
Pix2Pix-Zero	SD1.5 reproduced	DDIM	–	–	–	–	–	–	–	–
Pix2Pix-Zero	SD1.5 reproduced	DirectInv	–	**49.99 ± 51.39**	**21.47 ± 5.49**	**142.46 ± 115.06**	**133.49 ± 154.49**	**77.49 ± 15.64**	23.53 ± 4.40	21.09 ± 4.77
Pix2Pix-Zero	Best uniform CFG1	LoRA+DirectInv r16	8	51.63 ± 50.83	21.25 ± 5.48	147.07 ± 117.89	137.65 ± 153.16	76.94 ± 15.62	23.39 ± 4.35	20.95 ± 4.64
Pix2Pix-Zero	Best final-tail CFG1	LoRA+DirectInv r8	4	51.67 ± 51.06	21.33 ± 5.43	144.87 ± 117.21	135.70 ± 153.36	77.15 ± 15.58	**23.56 ± 4.29**	21.01 ± 4.67
Pix2Pix-Zero	Cycle loss ($\lambda = 1$)	LoRA r8	4	62.95 ± 47.81	20.32 ± 4.68	179.04 ± 114.37	148.97 ± 147.13	74.83 ± 15.52	22.88 ± 4.60	20.78 ± 4.85
Pix2Pix-Zero	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r8	4	51.19 ± 50.86	21.36 ± 5.71	145.33 ± 118.31	138.62 ± 156.68	77.12 ± 15.63	23.40 ± 4.43	20.97 ± 4.72
Pix2Pix-Zero	Cycle loss ($\lambda = 1$)	LoRA r16	8	63.73 ± 50.35	20.33 ± 4.75	179.45 ± 116.80	153.01 ± 165.31	74.68 ± 15.74	22.89 ± 4.50	20.73 ± 4.80
Pix2Pix-Zero	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r16	8	52.16 ± 52.08	21.22 ± 5.45	147.00 ± 121.01	140.33 ± 163.90	76.95 ± 15.75	23.43 ± 4.35	20.98 ± 4.74
Pix2Pix-Zero	Cycle loss ($\lambda = 1$)	LoRA r32	16	63.53 ± 49.50	20.29 ± 4.67	179.57 ± 114.54	151.14 ± 157.11	74.85 ± 15.43	22.89 ± 4.51	20.71 ± 4.81
Pix2Pix-Zero	Cycle loss ($\lambda = 1$)	LoRA+DirectInv r32	16	53.15 ± 55.55	21.24 ± 5.45	146.02 ± 118.51	138.96 ± 162.09	76.95 ± 15.60	23.45 ± 4.28	**21.11 ± 4.70**

Cycle loss lambda comparison

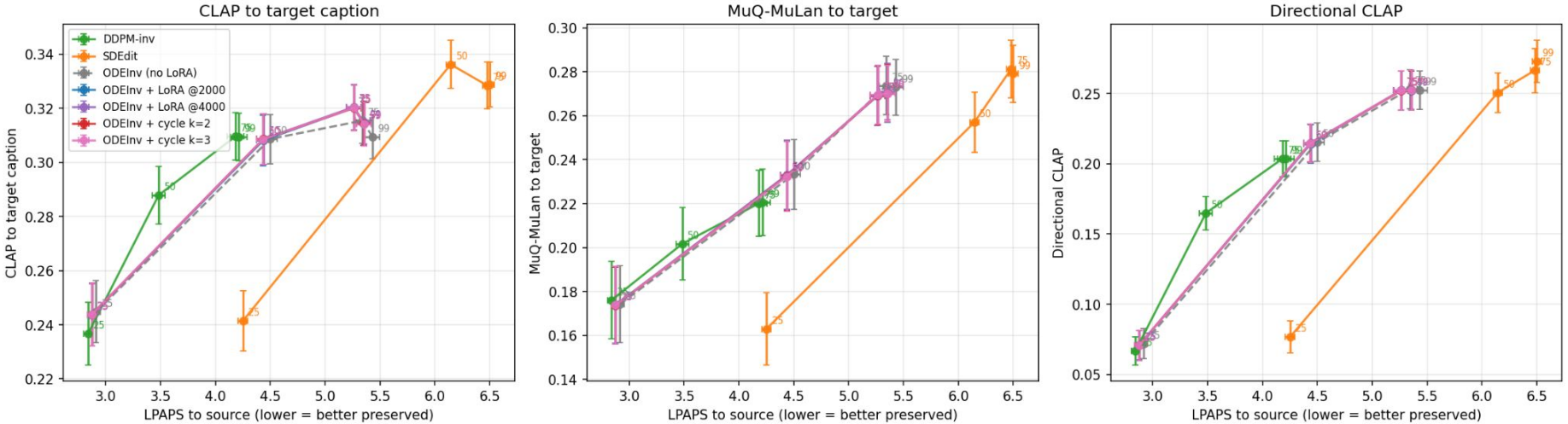
Editor	Setup	λ_{cycle}	Structure ($\times 10^3$) ↓	PSNR ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP edit ↑
P2P	Paper DirectInv	–	**11.65**	**27.22**	**54.55**	**32.86**	**84.76**	25.02	22.10
P2P	Cycle LoRA+DirectInv	0.5							
P2P	Cycle LoRA+DirectInv	1	14.45 ± 11.85	25.92 ± 5.39	74.51 ± 52.35	45.37 ± 46.89	83.37 ± 12.32	25.27 ± 3.84	22.31 ± 4.84
P2P	Cycle LoRA+DirectInv	2	14.41 ± 11.89	25.92 ± 5.28	74.10 ± 51.96	44.87 ± 46.24	83.39 ± 12.29	*25.32 ± 3.86*	*22.36 ± 4.86*
P2P	Cycle LoRA+DirectInv	4							
P2P	Cycle LoRA+DirectInv	8							
P2P	Cycle LoRA+DirectInv	16	*14.24 ± 11.75*	*26.01 ± 5.30*	*72.52 ± 50.57*	*43.61 ± 43.88*	*83.56 ± 12.26*	25.31 ± 3.89	22.31 ± 4.84
PnP	Paper DirectInv	–	**24.29**	**22.46**	**106.06**	**80.45**	79.68	25.41	22.62
PnP	Cycle LoRA+DirectInv	0.5							
PnP	Cycle LoRA+DirectInv	1	*24.38 ± 20.67*	22.36 ± 4.48	110.28 ± 62.53	81.87 ± 60.30	79.88 ± 12.81	25.81 ± 3.74	*22.83 ± 4.86*
PnP	Cycle LoRA+DirectInv	2	24.38 ± 20.67	22.36 ± 4.47	110.27 ± 62.51	81.87 ± 60.30	79.88 ± 12.81	25.81 ± 3.74	22.83 ± 4.86
PnP	Cycle LoRA+DirectInv	4	24.38 ± 20.70	22.36 ± 4.48	110.27 ± 62.54	81.87 ± 60.30	*79.88 ± 12.81*	25.81 ± 3.73	22.82 ± 4.86
PnP	Cycle LoRA+DirectInv	8	24.39 ± 20.74	*22.36 ± 4.48*	*110.26 ± 62.50*	*81.86 ± 60.30*	79.88 ± 12.81	*25.82 ± 3.74*	22.83 ± 4.86
PnP	Cycle LoRA+DirectInv	16							
MasaCtrl	Paper DirectInv	–	**24.70**	**22.64**	**87.94**	**81.09**	**81.33**	24.38	21.35
MasaCtrl	Cycle LoRA+DirectInv	0.5							
MasaCtrl	Cycle LoRA+DirectInv	1	28.49 ± 21.61	21.89 ± 4.89	98.91 ± 58.48	98.95 ± 84.46	80.50 ± 12.73	*24.63 ± 3.64*	21.48 ± 4.67
MasaCtrl	Cycle LoRA+DirectInv	2	28.53 ± 21.67	21.90 ± 4.92	98.83 ± 58.41	99.05 ± 84.65	80.50 ± 12.72	24.62 ± 3.63	21.50 ± 4.66
MasaCtrl	Cycle LoRA+DirectInv	4	28.51 ± 21.64	*21.90 ± 4.93*	98.89 ± 58.55	98.91 ± 84.48	80.50 ± 12.73	24.62 ± 3.64	*21.50 ± 4.67*
MasaCtrl	Cycle LoRA+DirectInv	8	*28.47 ± 21.61*	21.90 ± 4.92	*98.77 ± 58.31*	*98.91 ± 84.56*	*80.52 ± 12.71*	24.59 ± 3.65	21.47 ± 4.67
MasaCtrl	Cycle LoRA+DirectInv	16							
Pix2Pix-Zero	Paper DirectInv	–	**49.22**	**21.53**	**138.98**	**127.32**	**77.05**	23.31	21.05
Pix2Pix-Zero	Cycle LoRA+DirectInv	0.5							
Pix2Pix-Zero	Cycle LoRA+DirectInv	1	52.16 ± 52.08	21.22 ± 5.45	147.00 ± 121.01	140.33 ± 163.90	76.95 ± 15.75	*23.43 ± 4.35*	20.98 ± 4.74
Pix2Pix-Zero	Cycle LoRA+DirectInv	2	53.06 ± 53.93	21.25 ± 5.44	*146.22 ± 119.24*	*139.69 ± 165.86*	76.97 ± 15.66	23.33 ± 4.48	20.87 ± 4.73
Pix2Pix-Zero	Cycle LoRA+DirectInv	4	*52.06 ± 52.28*	*21.27 ± 5.54*	146.70 ± 120.31	139.76 ± 162.71	*77.02 ± 15.68*	23.41 ± 4.50	*21.08 ± 4.76*
Pix2Pix-Zero	Cycle LoRA+DirectInv	8							
Pix2Pix-Zero	Cycle LoRA+DirectInv	16							

SD1.4 results

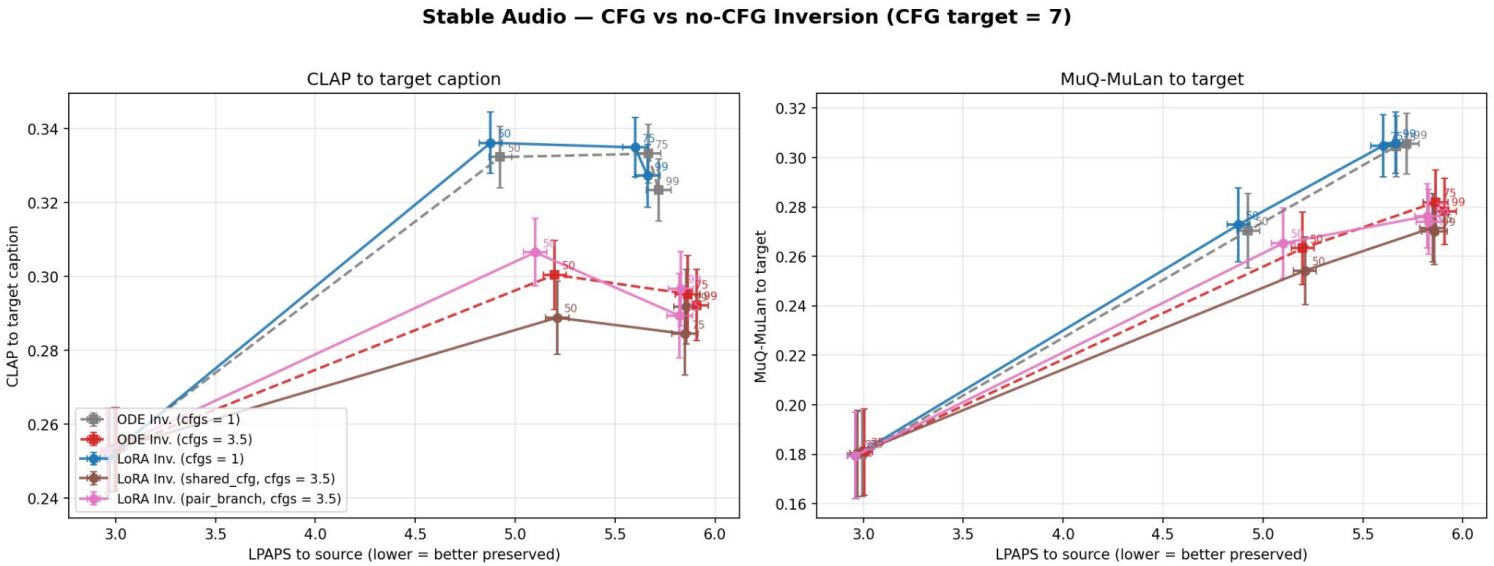
Editor	Training / source	Inversion	α	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP target ↑	CLIP target (edit) ↑
P2P	Paper original	DDIM	–	69.43	17.87	208.80	219.88	71.14	25.01	22.44
P2P	Paper original	DirectInv	–	**11.65**	**27.22**	**54.55**	**32.86**	**84.76**	25.02	22.10
P2P	Uniform	LoRA r8	4	69.42 ± 37.68	17.81 ± 4.09	209.60 ± 104.79	223.18 ± 147.73	71.36 ± 15.79	25.28 ± 3.79	22.65 ± 4.55
P2P	Uniform	LoRA+DirectInv r8	4	14.96 ± 11.73	25.98 ± 5.42	73.01 ± 51.04	44.75 ± 48.43	83.44 ± 12.32	**25.32 ± 3.81**	22.42 ± 4.81
P2P	Uniform	LoRA r16	8	69.57 ± 37.82	17.77 ± 3.96	210.24 ± 105.02	224.13 ± 147.89	71.30 ± 15.81	25.25 ± 3.77	22.64 ± 4.54
P2P	Uniform	LoRA+DirectInv r16	8	14.87 ± 11.44	26.01 ± 5.40	71.86 ± 47.55	43.58 ± 43.74	83.52 ± 12.03	25.27 ± 3.78	22.25 ± 4.80
P2P	Final-tail	LoRA r8	4	70.17 ± 40.73	17.78 ± 3.96	209.83 ± 105.38	223.76 ± 147.96	71.30 ± 15.79	25.31 ± 3.76	**22.67 ± 4.58**
P2P	Final-tail	LoRA+DirectInv r8	4	14.68 ± 11.29	26.03 ± 5.31	71.27 ± 46.80	42.87 ± 41.95	83.57 ± 12.00	25.28 ± 3.80	22.31 ± 4.81
P2P	Final-tail	LoRA r32	16	69.67 ± 39.71	17.81 ± 4.14	210.24 ± 105.17	224.12 ± 148.13	71.28 ± 15.85	25.28 ± 3.75	22.64 ± 4.61
P2P	Final-tail	LoRA+DirectInv r32	16	14.95 ± 11.45	25.93 ± 5.23	72.52 ± 48.90	43.65 ± 43.54	83.51 ± 12.05	25.31 ± 3.74	22.42 ± 4.80
MasaCtrl	Paper original	DDIM	–	28.38	22.17	106.62	86.97	79.67	23.96	21.16
MasaCtrl	Paper original	DirectInv	–	**24.70**	**22.64**	**87.94**	**81.09**	**81.33**	24.38	21.35
MasaCtrl	Uniform	LoRA r8	4	48.34 ± 27.65	20.01 ± 4.33	177.42 ± 91.98	139.74 ± 102.62	73.97 ± 15.36	20.69 ± 4.48	19.18 ± 4.66
MasaCtrl	Uniform	LoRA+DirectInv r8	4	28.83 ± 21.08	21.96 ± 4.89	98.38 ± 59.39	97.30 ± 83.39	80.44 ± 12.96	24.54 ± 3.68	21.48 ± 4.67
MasaCtrl	Uniform	LoRA r16	8	48.41 ± 27.87	19.99 ± 4.31	177.45 ± 91.92	140.18 ± 102.96	73.95 ± 15.34	20.65 ± 4.53	19.22 ± 4.69
MasaCtrl	Uniform	LoRA+DirectInv r16	8	28.94 ± 21.32	21.95 ± 4.89	98.39 ± 59.54	97.78 ± 84.24	80.43 ± 12.97	**24.58 ± 3.66**	**21.49 ± 4.64**
MasaCtrl	Final-tail	LoRA r8	4	48.13 ± 27.20	20.01 ± 4.34	175.59 ± 90.24	139.97 ± 103.39	74.16 ± 15.20	20.73 ± 4.62	19.25 ± 4.71
MasaCtrl	Final-tail	LoRA+DirectInv r8	4	28.85 ± 20.99	21.96 ± 4.91	97.84 ± 58.86	97.62 ± 83.80	80.49 ± 12.88	24.55 ± 3.61	21.45 ± 4.68
MasaCtrl	Final-tail	LoRA r32	16	48.06 ± 26.71	19.94 ± 4.27	178.10 ± 91.75	141.60 ± 105.64	73.94 ± 15.28	20.67 ± 4.47	19.23 ± 4.68
MasaCtrl	Final-tail	LoRA+DirectInv r32	16	28.73 ± 20.78	21.96 ± 4.91	98.38 ± 59.29	97.71 ± 84.72	80.43 ± 12.92	24.51 ± 3.59	21.46 ± 4.65

$$\mathcal{L}_{\text{cycle}}^{(k)} = \left\| z_{i+1} - \underbrace{F_\theta^{\text{gen}} \circ \cdots \circ F_\theta^{\text{gen}}}_{k} \left(\underbrace{F_\phi^{\text{inv}} \circ \cdots \circ F_\phi^{\text{inv}}}_{k} (z_{i+1}) \right) \right\|^2$$

Stable Audio Open, every method at one operating point (cfg_tar 3.5, 100-step grid, 115 edits) — points labelled by tstart



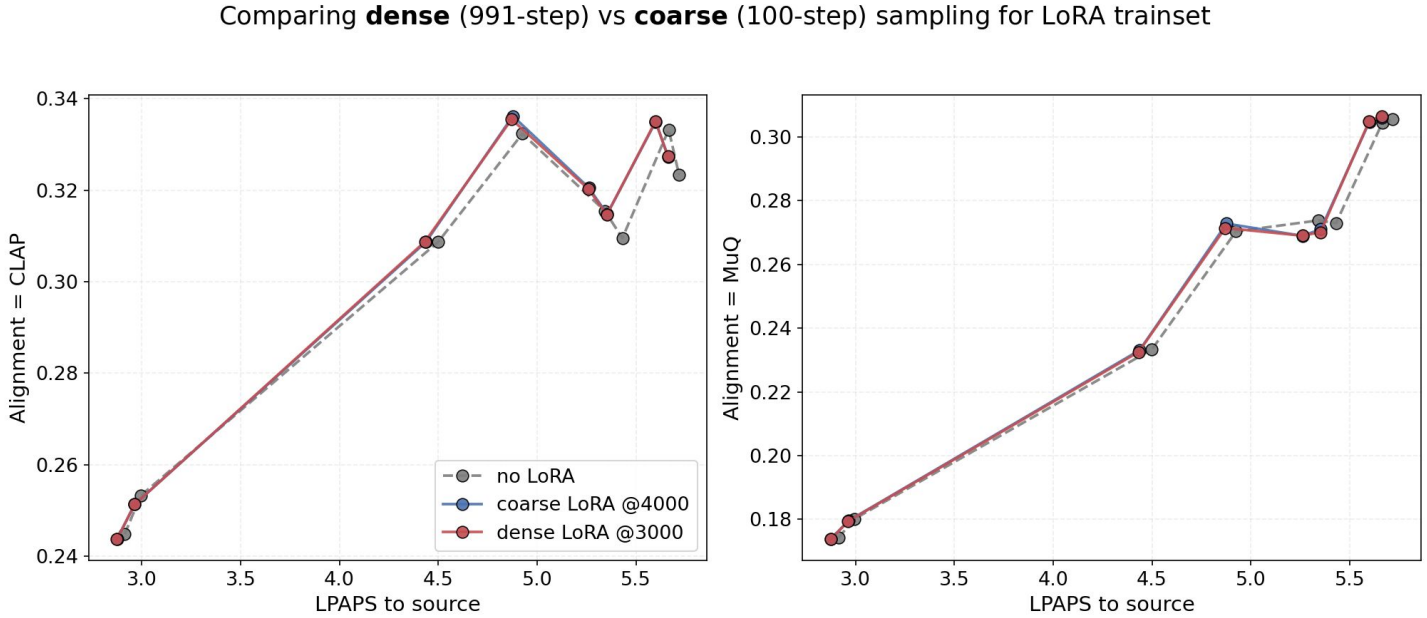
Stable Audio Open – train LoRA on higher guidance scale



Results:

1) Inversion with higher guidance scale does not work

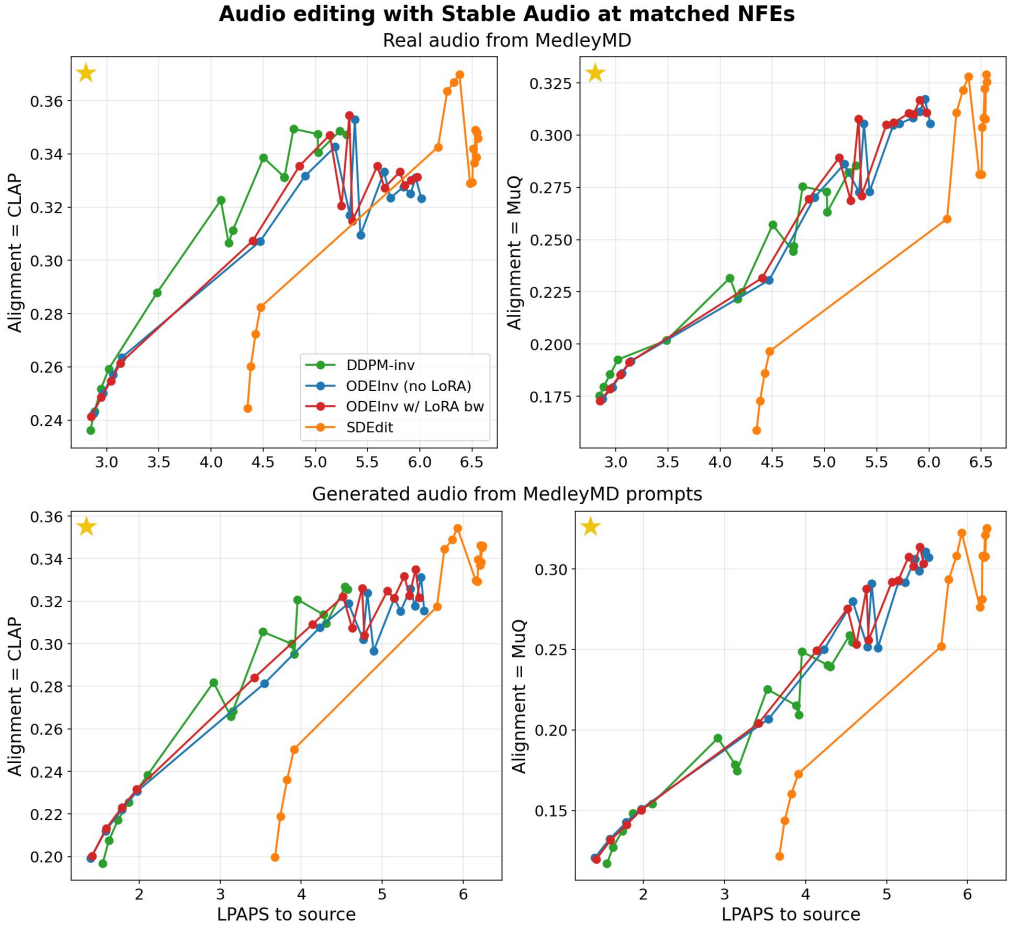
Stable Audio Open – dense vs coarse sampling



Results:

1) Training LoRA on higher quality **generations** does not yield significant difference

Stable Audio Open – editing real vs generated audio



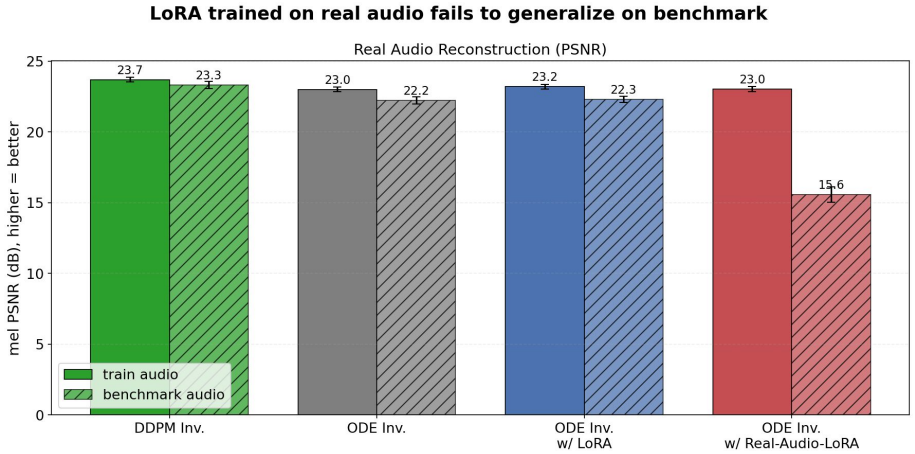
Results:

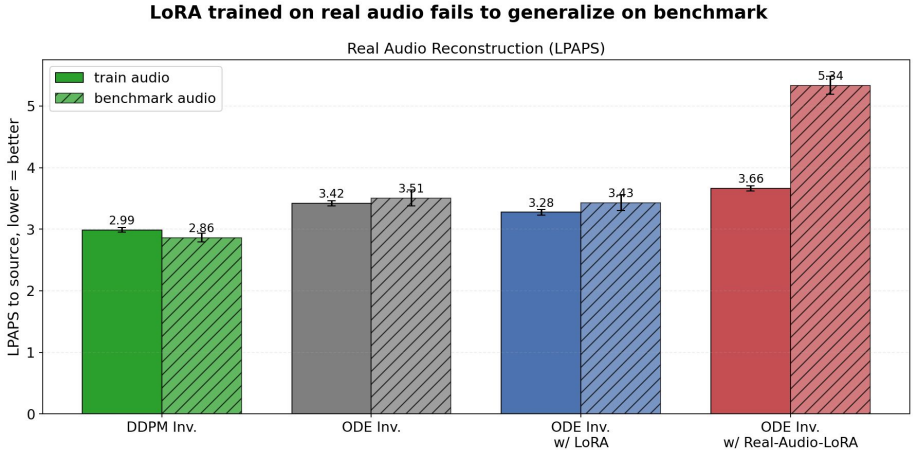
1) DDPM Inv. cannot reach MuQ as high as ODE Inv
2) Gap between DDPM and ODE Inv is much smaller for generated audio

Follow-up experiment:

1) Train LoRA on MusicCaps real audio by collecting DM preds from forward-diffused music

Stable Audio Open – training LoRA on real audios





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Results:

(R1) LoRA trained on real audio reconstructs **slightly** worse than ODE Inv. / ODE Inv.+LoRA even at training dataset

(R2) It completely fails to generalize to benchmark audio (not seen during training)

My opinion:

(O1) (R1) seems to be really weird

(O2) rationale for (R2): musiccaps (trainset) has 10-sec audios only, while benchmark dataset (MedleyMD) has longer tracks

Potential follow-up exp:

(E1) Train on the subset of MedleyMD to check if (O2) is True

Mix data 80% recap COCO + 20% diffusion DB artificial prompt

Editor	Inversion	Source	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑	CLIP whole ↑	CLIP edited ↑
P2P	Direct Inversion	SD1.5 reproduction	**11.01**	**27.27**	**54.01**	**32.60**	**85.32**	25.20	22.14
P2P	LoRA	Mixture r16	70.38	17.82	210.82	220.69	71.36	**25.40**	**22.62**
P2P	LoRA + DirectInv	Mixture r16	14.29	25.96	73.21	43.92	83.43	25.26	22.31
MasaCtrl	Direct Inversion	SD1.5 reproduction	**23.70**	**22.67**	**87.85**	**80.85**	**81.85**	24.60	21.46
MasaCtrl	LoRA	Mixture r16	49.29	19.92	180.08	142.58	73.89	20.61	19.18
MasaCtrl	LoRA + DirectInv	Mixture r16	28.46	21.90	98.86	98.75	80.50	**24.62**	**21.51**
Pix2Pix-Zero	Direct Inversion	SD1.5 reproduction	**49.09**	**21.57**	**138.39**	**129.77**	**77.73**	**23.47**	**21.08**
Pix2Pix-Zero	LoRA	Mixture r16	63.12	20.28	180.40	153.79	74.61	22.93	20.73
Pix2Pix-Zero	LoRA + DirectInv	Mixture r16	51.50	21.29	146.32	137.22	77.00	23.41	20.98
PnP	Direct Inversion	SD1.5 reproduction	**23.35**	**22.46**	**105.49**	**79.93**	**80.22**	25.64	22.67
PnP	LoRA	Mixture r16	26.93	22.31	113.28	82.54	79.56	25.70	22.79
PnP	LoRA + DirectInv	Mixture r16	24.37	22.36	110.31	81.89	79.88	**25.80**	**22.84**

Analiza najgorszych przykladow

Editor	Artificial worst / remaining	Luminance contrast z	Edge detail z	Unedited area worst / remaining
Pix2Pix-Zero	28% / 39%	+0.10	+0.41	85% / 67%
MasaCtrl	52% / 38%	+0.75	+0.53	84% / 67%
PnP	68% / 37%	+0.16	+0.78	83% / 67%

Metryki rekonstrukcji na nieedytowanych obrazach

Editor	Configuration	Adapter	Structure ($\times 10^3$) ↓	PSNR (dB) ↑	LPIPS ($\times 10^3$) ↓	MSE ($\times 10^4$) ↓	SSIM ($\times 10^2$) ↑
P2P	DirectInv	–	38.79 ± 26.95	19.65 ± 3.98	158.91 ± 89.79	154.93 ± 128.85	65.52 ± 14.98
P2P	Best uniform	LoRA+DirectInv r8	**3.31 ± 1.79**	**27.28 ± 3.93**	37.17 ± 14.27	**28.19 ± 30.29**	79.99 ± 12.45
P2P	Best final-tail	LoRA+DirectInv r8	3.31 ± 1.81	27.28 ± 3.93	37.17 ± 14.27	28.19 ± 30.29	79.99 ± 12.45
P2P	Cycle loss	LoRA+DirectInv r8	3.31 ± 1.82	27.28 ± 3.93	**37.15 ± 14.25**	28.19 ± 30.28	**79.99 ± 12.45**
PnP	DirectInv	–	3.38 ± 1.85	**27.26 ± 3.92**	37.15 ± 14.23	**28.24 ± 30.31**	79.97 ± 12.44
PnP	Best uniform	LoRA+DirectInv r8	3.37 ± 1.84	27.26 ± 3.92	**37.13 ± 14.21**	28.24 ± 30.31	**79.97 ± 12.44**
PnP	Best final-tail	LoRA+DirectInv r32	3.37 ± 1.82	27.26 ± 3.92	37.15 ± 14.23	28.24 ± 30.31	79.97 ± 12.44
PnP	Cycle loss	LoRA+DirectInv r8	**3.37 ± 1.83**	27.26 ± 3.92	37.15 ± 14.25	28.24 ± 30.31	**79.97 ± 12.44**
MasaCtrl	DirectInv	–	**3.31 ± 1.84**	**27.28 ± 3.93**	**37.15 ± 14.26**	**28.19 ± 30.28**	**79.99 ± 12.45**
MasaCtrl	Best uniform	LoRA+DirectInv r16	3.69 ± 3.72	27.22 ± 3.92	37.84 ± 15.70	28.50 ± 30.49	79.91 ± 12.46
MasaCtrl	Best final-tail	LoRA+DirectInv r32	3.75 ± 3.87	27.21 ± 3.90	37.95 ± 15.79	28.54 ± 30.52	79.89 ± 12.48
MasaCtrl	Cycle loss	LoRA+DirectInv r8	3.63 ± 3.03	27.22 ± 3.91	37.84 ± 15.29	28.41 ± 30.29	79.91 ± 12.44
Pix2Pix-Zero	DirectInv	–	**3.35 ± 1.83**	**27.36 ± 4.00**	37.11 ± 14.36	**27.96 ± 30.29**	**80.05 ± 12.46**
Pix2Pix-Zero	Best uniform	LoRA+DirectInv r16	3.37 ± 1.89	27.36 ± 4.00	**37.11 ± 14.36**	27.96 ± 30.29	80.05 ± 12.46
Pix2Pix-Zero	Best final-tail	LoRA+DirectInv r8	3.36 ± 1.87	27.36 ± 4.00	37.11 ± 14.36	27.96 ± 30.28	80.05 ± 12.45
Pix2Pix-Zero	Cycle loss	LoRA+DirectInv r8	3.36 ± 1.89	27.36 ± 4.00	**37.10 ± 14.35**	27.96 ± 30.29	**80.05 ± 12.46**